BN-100 LUX

PART III

Chemical, Physical, Mechanical and Biological Information

Medical Device Homologation Dossier

SKALBEL SANTE

Contents

PART III: Chemical, Physical, Mechanical and Biological Information **2**
3.1 Body of Data . . . 2
3.1.1 Scope of the Data . . . 2
3.1.2 Technical Information Covered . . . 2
3.1.3 Applicability to the BN-100 LUX . . . 3
3.1.4 General Information . . . 3
3.1.5 Manufacturing . . . 8
3.1.6 Control of Raw Material . . . 20
3.2 Finished Product . . . 25
3.2.1 Description and Composition of the Finished Product . . . 25
3.2.2 Manufacturing . . . 28
3.2.3 Control of the Finished Product . . . 39
3.2.4 Reference Standards . . . 45
3.2.5 Stability . . . 47
3.3 Risk Analysis Report . . . 49
3.3.1 Risk Analysis . . . 49
3.3.2 Risk Evaluation . . . 53
3.3.3 Risk Control . . . 55
3.3.4 Overall Residual Risk Evaluation . . . 59
3.4 Annexes . . . 62
3.4.1 General . . . 62
3.4.2 Manufacturer Documentation . . . 62
3.4.3 Technical and Safety Documentation . . . 63
3.4.4 Device Identification Documentation . . . 63
3.4.5 Risk Management Annexes . . . 64
3.4.6 Supporting Evidence . . . 64
3.4.7 Document Control . . . 64
3.4.8 Documentary Basis . . . 65
3.5 Bibliographical References . . . 65
3.5.1 General . . . 65
3.5.2 Manufacturer Documentation . . . 65
3.5.3 Applicable Standards and Reference Documents . . . 65
3.5.4 Use of References . . . 66
3.5.5 Document Identification . . . 66
3.5.6 Documentary Basis . . . 66
Manufacturer's Component Certificates . . . 67

PART III: Chemical, Physical, Mechanical and Biological Information

3.1 Body of Data

3.1.1 Scope of the Data

This section presents the technical information supporting the characterization of the **BN-100 LUX** medical device for the purpose of the medical device registration and homologation dossier.

The **BN-100 LUX** is a reusable neonatal radiant warmer intended to provide controlled thermal support to newborns and neonates through non-contact radiant heating.

The information presented in Part III addresses the physical, mechanical, electrical, electronic, material, manufacturing, performance, safety and risk-management characteristics applicable to the device.

The information is structured according to the applicable requirements of the homologation dossier and is adapted to the nature and technological characteristics of the **BN-100 LUX** as an active electrical medical device.

3.1.2 Technical Information Covered

The technical information presented in the following sections covers, as applicable:

- identification and designation of the device;
- general physical and functional characteristics;
- materials, components and accessories;
- device configuration and principal operating functions;
- manufacturing processes and manufacturing controls;
- controls applied to materials, components and critical manufacturing steps;
- specifications and characteristics applicable to the finished device;
- applicable standards and technical references;
- environmental, service-life and maintenance-related characteristics;
- risk-management and risk-control information available for the device; and
- supporting technical documentation and references.

The level of detail provided for each subject is determined by its applicability to the **BN-100 LUX**, the nature of the information concerned and the supporting manufacturer documentation available for the device.

3.1.3 Applicability to the BN-100 LUX

The applicability of each technical information category is assessed according to the nature, intended purpose, technological characteristics and clinical use of the **BN-100 LUX**.

Information categories applicable to medical devices are addressed according to the corresponding characteristics and functions of the device.

Requirements or information categories that are not applicable to the **BN-100 LUX** are not presented as device characteristics.

Where a requirement is determined to be not applicable, its non-applicability is identified together with an appropriate technical justification.

This approach ensures that the dossier reflects the actual characteristics of the device while avoiding the introduction of unsupported information.

3.1.4 General Information

Medical Purpose

The BN-100 LUX is intended to provide controlled thermal support to newborns and neonates requiring a suitable warming environment.

Its intended clinical use includes neonatal and pediatric care areas and delivery rooms, according to applicable clinical procedures and institutional practices.

Intended Users

The device is intended to be operated by appropriately trained healthcare professionals working in relevant neonatal, pediatric and delivery-care environments.

The operator is responsible for selecting the appropriate operating mode and monitoring the infant during use.

Target Patient Population

The intended patient population consists of newborns and neonates requiring thermal support.

Operating Principle

The BN-100 LUX provides thermal energy through infrared radiant heating. The heating source generates infrared radiation directed toward the infant through a reflector assembly.

The thermal support provided by the device is controlled through the device control system and may be monitored using the skin temperature sensor.

Device Configuration

The principal configuration identified in the manufacturer documentation includes the radiant heating system, control system, infant bed, structural support system, and associated mechanical and electrical components.

Depending on the supplied configuration, additional accessories or functional modules may be provided, including:

- skin temperature sensor;
- LED phototherapy unit;
- spotlight;
- X-ray cassette tray;
- intravenous support pole or hook;
- oxygen cylinder bracket;
- low-pressure suction equipment; and
- other accessories identified by the manufacturer.

Designation

Commercial and Technical Identification The medical device covered by this dossier is the **BN-100 LUX neonatal radiant warmer**.

The manufacturer identified in the available documentation is:

Zhengzhou Dison Instrument and Meter Co., Ltd.
No. 67, Jinsuo Road, High-Tech Zone Development,
Zhengzhou, Henan, China, 450001.

Device Designation

- **Device name:** Neonatal Radiant Warmer
- **Model:** BN-100 LUX
- **Product designation:** BN-100 LUX
- **Device type:** Active electrical medical device

Nature of the Device The BN-100 LUX is a reusable medical device. It is not a medicinal product and does not contain an active pharmaceutical substance.

General Properties

General Description The BN-100 LUX is an active electrical neonatal radiant warmer designed to provide thermal support to newborns and neonates.

The device uses non-contact infrared radiant heating. The heating source is positioned above the infant bed, and a reflector assembly directs radiant energy toward the infant. The control system supports management of the heating function and monitoring of skin temperature when the corresponding sensor and operating mode are used.

The detailed operating characteristics and limitations are described in the following subsections.

Heating and Temperature Monitoring According to the manufacturer documentation, the principal heating and temperature-monitoring characteristics include:

- heating tube rated power: **800 W**;
- skin temperature control range: 32–38 °C;
- stated overheating level 1: 39 °C;
- stated overheating level 2: 40 °C;
- stated temperature deviation alarm: ±1 °C;
- stated skin temperature sensor precision: ≤ 0.3 °C.

The heating mechanism is non-contact. Thermal energy is transferred from the heating source toward the infant primarily through radiant heat transfer.

The skin temperature sensor provides temperature information to the control system and supports temperature monitoring and automatic thermal control when the corresponding operating mode is selected.

Correct sensor placement and attachment are important for reliable temperature monitoring. The sensor shall be used in accordance with the manufacturer's instructions for use, including the specified precautions concerning incorrect placement or detachment.

The stated temperature thresholds, deviation alarm and sensor precision shall be interpreted together with the applicable manufacturer specifications and technical documentation.

Control and Alarm System The device incorporates electronic control circuitry associated with heating, temperature monitoring and alarm functions.

The available manufacturer documentation identifies control circuits, low-voltage circuits, relay and alarm circuitry, and heating-control switching components. Detailed circuit diagrams, component specifications, electrical tolerances and safety-related circuit analyses are not reproduced in this subsection.

According to the manufacturer documentation, the specified alarm conditions include:

- mains power failure;
- skin temperature sensor failure;
- overheating; and
- temperature deviation.

The documentation also identifies an internal battery associated with the alarm function during mains power interruption. The battery configuration and the operation of the alarm during a power failure are described in the applicable manufacturer documentation.

Alarm functions, operating modes and associated precautions shall be interpreted in accordance with the instructions for use and the relevant technical documentation.

Electrical Characteristics and Safety Classification According to the manufacturer documentation, the principal electrical characteristics are:

- rated supply voltage: **220 V** ±22 **V AC**;
- rated frequency: **50 Hz** ±1 **Hz**;
- maximum input power: **1000 VA**.

The manufacturer documentation identifies the following electrical safety and operating characteristics:

- protection against electric shock: **Class I**;
- applied part classification: **Type BF**;
- ingress protection rating: **IPX0**;
- mode of operation: **Continuous operation**.

The manufacturer also states that the device is not intended for use in the presence of flammable anaesthetic mixtures with air, oxygen or nitrous oxide.

These characteristics concern electrical safety and operating conditions. They shall not be confused with the regulatory classification applicable to the Algerian homologation process.

Applicable electrical safety requirements and conformity documentation are addressed in the relevant technical sections of the dossier.

Environmental Operating Conditions The specified environmental conditions for operation are:

- **Ambient temperature:** +18 °C to +30 °C;
- **Relative humidity:** 30% to 75%;
- **Atmospheric pressure:** 700 to 1060 hPa;
- **Air velocity:** less than 0.3 m/s.

The device should be operated within the specified environmental limits to maintain its intended performance and temperature measurement characteristics.

Storage and Transportation Conditions The specified storage and transportation conditions are:

- **Temperature:** −40 °C to +55 °C;
- **Relative humidity:** not exceeding 95%;
- **Atmospheric pressure:** 500 to 1060 hPa.

The device should be protected against environmental conditions that could adversely affect its mechanical, electrical or electronic components during storage and transportation.

Mechanical Characteristics and Load Limits The BN-100 LUX incorporates an infant bed and auxiliary tray structures with specified load limits.

According to the manufacturer documentation:

- maximum infant bed load: **100 N**, corresponding approximately to **10 kg**;
- maximum tray load: **20 N**, corresponding approximately to **2 kg**.

These limits apply to their respective structural components and shall not be interpreted as interchangeable load ratings.

Service Life The specified expected service life of the BN-100 LUX is **8 years**. The specified service life of the heating tube is **25,000 hours**.

These service-life specifications do not eliminate the need for preventive maintenance, inspection, cleaning, functional verification and component replacement where required.

Actual operational life depends on the conditions of use, maintenance practices and the condition of safety-critical components.

Reusability and Cleaning The BN-100 LUX is a reusable medical device and is not intended to be supplied as a sterile, single-use medical device.

Cleaning and disinfection shall be performed in accordance with the manufacturer's instructions for use and applicable institutional procedures.

Cleaning and disinfection agents shall be selected with due consideration for their compatibility with the device materials and surfaces.

Patient Safety-Related Characteristics The device incorporates operational precautions intended to reduce risks associated with neonatal radiant heating.

The manufacturer's safety information includes:

- warnings concerning excessive heating and potential burns;
- precautions related to incorrect placement or detachment of the skin temperature sensor;
- precautions concerning oxygen-enriched environments and fire hazards;
- precautions associated with manual heating modes; and
- precautions associated with optional phototherapy functions, including protection of the infant's eyes.

These precautions shall be considered together with the manufacturer's instructions for use and risk-management documentation.

Material Information The manufacturer documentation identifies the following materials for specific components:

- carbon steel for the radiant heating tube;
- Lucite/Plexiglas for the side baffles; and
- polycarbonate for the suction bottle.

These material identifications apply to the components specified in the manufacturer documentation and do not constitute a complete material declaration for the entire device.

Applicable Standards and Verification Evidence The manufacturer documentation identifies the following standards and technical requirements in relation to the device:

- IEC 60601-1;
- IEC 60601-1-8;
- IEC 60601-2-21;
- CISPR 11, Group 1, Class A;
- IEC 61000-4-2;
- IEC 61000-4-3;
- IEC 61000-4-4; and
- IEC 61000-4-5.

These references provide the technical framework for the relevant safety, alarm, electromagnetic compatibility and radiant-warmer requirements. The associated conformity documentation and test evidence are addressed in the relevant technical sections of the dossier.

3.1.5 Manufacturing

Manufacturer(s)

Manufacturer Identification The medical device covered by this application is the **BN-100 LUX** Infant Radiant Warmer.

The manufacturer identified in the available technical documentation and quality-management-system documentation is:

- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.
- **Address:** No. 67, Jinsuo Road, High-Tech Development Zone, Zhengzhou, Henan, China
- **Postal Code:** 450001

The available manufacturer documentation identifies Zhengzhou Dison Instrument and Meter Co., Ltd. as the manufacturer responsible for the device.

Manufacturing Site Information The address stated above is the manufacturer address identified in the available documentation.

No separate manufacturing-site address, additional manufacturing facility, subcontractor or other manufacturer is identified in the documentation considered in this section. Accordingly, no additional manufacturing entity or site is described here.

Quality Management System Certification The manufacturer holds an ISO 13485:2016 quality-management-system certification issued by BSI. The certificate information available for this submission is summarized below:

- **Certificate Holder:** Zhengzhou Dison Instrument and Meter Co., Ltd.
- **Standard:** ISO 13485:2016
- **Certificate No.:** MD 796665
- **Certification Body:** BSI
- **Original Registration Date:** 05 March 2024
- **Effective Date:** 05 March 2024
- **Expiry Date:** 04 March 2027

The certified scope includes the design and development, manufacture and distribution of the following medical devices:

- Baby Incubators;
- Infant Radiant Warmers;
- Infant Phototherapy Units;
- Transport Infant Incubators; and
- Reusable Pneumatic Pulmonary Resuscitators.

The ISO 13485:2016 certificate is included in the regulatory dossier as supporting documentation concerning the manufacturer's quality-management system.

EU MDR Quality Management System Certificate The manufacturer documentation also includes the BSI EU Quality Management System Certificate under Regulation (EU) 2017/745 (MDR), identified as Certificate No. MDR 796664 R000.

The certificate information is summarized as follows:

- **Certificate Holder:** Zhengzhou Dison Instrument and Meter Co., Ltd.
- **Certification Body:** BSI
- **Notified Body Number:** 2797
- **Regulatory Framework:** Regulation (EU) 2017/745
- **Certificate Issue Date:** 22 August 2025
- **Certificate Expiry Date:** 21 August 2030
- **Assessment Framework:** Annex IX of Regulation (EU) 2017/745

The certificate scope includes Infant Radiant Warmers classified as Class IIb within the scope specified in the certificate.

This certificate is included as supporting regulatory documentation concerning the manufacturer's quality-management system under the EU MDR framework. It is considered alongside the ISO 13485:2016 certificate and the other applicable technical and conformity documentation in this dossier.

Relationship to the BN-100 LUX The manufacturer's quality-management-system documentation covers activities associated with Infant Radiant Warmers. The BN-100 LUX is identified in this dossier as an Infant Radiant Warmer.

The ISO 13485:2016 certificate and the EU MDR quality-management system certificate provide complementary supporting information concerning the manufacturer's quality-management system and the activities within their respective certified scopes.

These certificates shall be considered together with the device's technical documentation, declarations of conformity, applicable standards and test reports included in the regulatory dossier.

Documentary Basis The manufacturer identification and address presented in this section are based on the available manufacturer documentation. The ISO 13485:2016 information is based on BSI Certificate No. MD 796665. The EU MDR quality-management-system information is based on BSI Certificate No. MDR 796664 R000.

Description of the Manufacturing Process and In-Process Controls

General Description The **BN-100 LUX** Infant Radiant Warmer is manufactured and assembled by Zhengzhou Dison Instrument and Meter Co., Ltd.

The available manufacturer documentation describes the principal structural assemblies, radiant heating assembly, operating and control assembly, temperature monitoring system, and associated electrical and electronic components.

The manufacturer documentation also indicates that production and post-production activities are conducted within the manufacturer's quality-management and risk-management framework.

However, the detailed controlled manufacturing procedures, individual work instructions, process parameters, inspection criteria and production records have not been identified in the documentation currently available for this submission.

Accordingly, this section describes only the manufacturing and assembly activities supported by the available manufacturer documentation and does not reconstruct undocumented production procedures.

Main Structural Base According to the available manufacturer documentation, the base structure incorporates aluminium alloy extrusions.

The documented manufacturing operations associated with the structural components include processing, welding and surface spraying.

The base may be equipped with four mobile castors with braking mechanisms, depending on the applicable device configuration.

These structural components provide the supporting framework for the infant crib, operating section and associated device assemblies.

The available documentation does not specify detailed process parameters, inspection criteria or acceptance limits for these manufacturing operations.

Infant Crib Assembly The infant crib incorporates steel structural components.

According to the manufacturer documentation, the structural components undergo forming, bending and welding operations followed by surface finishing.

The crib incorporates an X-ray cassette arrangement and transparent protective side panels.

The crib structure also incorporates a mechanism allowing adjustment of the inclination of the infant support surface.

The available documentation does not provide detailed manufacturing work instructions, dimensional tolerances or intermediate-product acceptance criteria for the crib assembly.

Radiant Heating Assembly The radiant heating assembly constitutes a safety- and performance-relevant assembly of the BN-100 LUX.

The applicable manufacturer documentation identifies structural components associated with the radiant heating assembly and describes the integration of the radiant heating element and reflector.

The manufacturer risk-management documentation identifies the air heater as a critical component.

The available documentation describes a fixed infrared heating element and associated protective components.

The radiant heating assembly also incorporates an adjustable-angle LED white light for illumination of the infant care area.

The available documentation does not provide a complete detailed production sequence, process parameters or specific intermediate acceptance criteria for the radiant heating assembly.

Accordingly, no additional manufacturing procedure or acceptance criterion is introduced in this dossier.

Operating and Control Assembly The operating and control assembly incorporates processed structural components together with the electrical and electronic components required for operation of the device.

The manufacturer documentation identifies electrical components including power transformers and control circuit boards.

The assembly also provides mounting and support for applicable accessories, such as the infusion hook and carrier tray, depending on the device configuration.

The available documentation does not provide the complete assembly work instructions or detailed production-control criteria for this assembly.

Temperature Monitoring and Alarm-Related Assembly The BN-100 LUX incorporates a skin temperature monitoring system and associated alarm functions.

The skin temperature sensor is connected to the control system and is used to monitor the infant's skin temperature.

The manufacturer risk-management documentation identifies the skin temperature sensor as a critical component and identifies safety functions associated with sensor failure, including short-circuit and open-circuit alarm conditions.

The assembly of the temperature monitoring and alarm-related components is therefore considered safety-relevant.

However, the available documentation does not provide separate production-stage test procedures or acceptance records for these functions.

No undocumented production test method or acceptance criterion is therefore introduced in this section.

Assembly and In-Process Controls The manufacturing process includes, as documented, the processing and assembly of mechanical structures, integration of the radiant heating assembly,

installation of electrical and electronic components, integration of temperature monitoring and alarm-related components, and assembly of applicable accessories and protective components.

The manufacturer's risk-management documentation indicates that manufacturing-process-related risks are considered within the overall risk-management process.

Production and post-production activities are required to follow the manufacturer's applicable processes, methods and control measures.

Particular attention is therefore relevant to assemblies and components associated with:

- radiant heating;
- skin temperature monitoring;
- electrical protection;
- alarm functions; and
- other safety-related device functions.

The available documentation does not provide a complete set of in-process inspection procedures, acceptance criteria or inspection records for each manufacturing stage.

Production and Post-Production Control The manufacturer's Risk Management Report identifies production and post-production activities as integral elements of the overall risk-management process. These activities follow the applicable processes, methods and measures established by the manufacturer and include product change control and customer feedback control.

The manufacturing and quality-management framework for the BN-100 LUX Infant Radiant Warmer is documented through the applicable manufacturer documentation, including the Risk Management Report, the User Manual and the ISO 13485:2016 quality-management-system certificate, where relevant. These documents provide the documentary basis for considering manufacturing, component-control and post-production risks within the device's overall safety and performance framework. The original manufacturer documentation is retained as supporting evidence in the regulatory dossier.

Control of Raw Materials and Components

General The **BN-100 LUX** Infant Radiant Warmer incorporates mechanical, electrical, electronic and patient-contact components relevant to the safety and performance of the device.

The manufacturer identifies critical components and relevant materials within the applicable product risk-management documentation. Their selection, specification and control form part of the manufacturer's product design, manufacturing, quality-management and risk-management processes.

The available manufacturer documentation provides information concerning selected component models, technical specifications and applicable standards, certifications or registration references.

Patient-Contact Materials The manufacturer's Risk Management Report identifies the following materials as having direct contact with the infant:

- **Baby mattress:** cotton material. The manufacturer identifies contact with the infant's skin for more than 24 hours within a 30-day period.
- **Skin temperature sensor probe:** stainless steel. The manufacturer identifies contact with the infant's skin for more than 24 hours within a 30-day period.

These patient-contact components are included in the manufacturer's risk-management assessment in relation to the intended use of the device. The applicable material information and patient-contact characteristics are documented in the manufacturer's risk-management documentation.

Critical Components The manufacturer identifies the following components as critical components of the **BN-100 LUX** within the applicable risk-management documentation.

No.	Component	Model / Specification	Standard / Certification Reference
1	Dual fuse input socket	SP-861	GB/T 17465.1-2022; GB/T 17465.6-2022; CCC No. 2003010204037238
2	Power supply cord	3VTJ216A250V / 3VTJ A10A2m	DIN VDE 0620-1 (VDE 0620-1):2010-02; VDE No. 40012265
3	Auxiliary socket outlets	ST-A02	EN 60320-1:2015; UL No. E230335
4	Relay	HF115F012-1HS3	GB/T 21711.1-2008; CQC No. CQC17002168381
5	Rocker switch	KCD2	ANSI/UL 61058-1; UL No. E239917
6	Fuse	F5AL250V	EN 60127-1:2006+A1:2011+A2:2015; HTT No. HTT180102295L
7	Rechargeable battery	B80H7A2H	IEC 62133-1:2017; Certificate No. DGSJH2023041902
8	Optocoupler	MOC3083	DIN EN 60747-5; VDE No. 102497
9	PCB base material	NPG-180BH	GB/T 4725-1992; GB/T 6040-2019; GB/T 19466.1-2004; GB/T 19466.2-2004; GB/T 19466.3-2004; ISO 11358-1:2014; CQC No. CQC21134297754
10	Internal wire	0.5 / 1.0 / 2.5 / 4.0 mm^2	GB/T 5023.5-2008 / IEC 60227-5:2003; JB/T 8734.3-2016; CCC No. 2002010105016822
11	Skin temperature sensor	T10-AS	MDD Annex II, excluding Section 4; TUV No. G1 088955 0007
12	Air heater	EQ-076-A	IEC 60601-1; manufacturer documentation indicates testing with equipment
13	Thyristor (TRIAC)	BTA12-600B	IEC 60601-1; manufacturer documentation indicates testing with equipment
14	Audible alarm	SFM-27	IEC 60601-1; manufacturer documentation indicates testing with equipment

No.	Component	Model / Specification	Standard / Certification Reference
15	Switching power supply	RPS-45-12	BS EN 60601-1:2006+A12:2014; TUV No. TA50328876
16	EMI filter	RV410-6	EN 60939-1:2010; EN 60939-2:2005; UDEM: M.2022.206.C74237

Control Basis The critical components identified above are addressed within the manufacturer's product design, manufacturing, quality-management and risk-management activities.

The manufacturer documentation identifies applicable technical standards, component specifications and, where applicable, certification or registration references associated with the listed components.

Detailed incoming inspection criteria, supplier qualification requirements, acceptance limits, inspection procedures and inspection records are controlled manufacturer documents. These detailed controls are not reproduced here where they are not included in the supporting documentation available for this submission.

Relation to Risk Management The identification of critical components is integrated into the manufacturer's risk-management process. Components associated with heating, temperature measurement, electrical power, electrical protection, alarm functions and patient contact are considered in the assessment and control of risks associated with the device.

The manufacturer's Risk Management Report, Document No. YF-7.3-24-02 (w), Version A/0, provides the primary documentary basis for the identification of the critical components and patient-contact materials described in this section.

Control of Critical Steps and Intermediate Products

General The manufacture and assembly of the **BN-100 LUX** Infant Radiant Warmer involve mechanical, electrical, electronic and functional assembly activities relevant to the safety and performance of the finished device.

The manufacturer considers risks associated with the device throughout its design, manufacturing and post-production lifecycle. Critical assemblies, safety-related functions and manufacturing- process risks are addressed within the manufacturer's quality-management and risk-management framework.

The applicable manufacturer documentation provides the basis for identifying the critical assemblies and functions described below.

Mechanical Assembly Controls The principal mechanical assemblies include the base structure, infant crib, radiant heating support structure and operating assembly.

The available manufacturer documentation identifies processing, welding, surface finishing and assembly activities associated with the relevant structural components.

Particular attention is relevant to the integration of structural components, protective components and movable elements that contribute to the safe operation of the device.

The infant crib assembly includes the infant support structure, X-ray cassette arrangement, transparent protective side panels and the mechanism for adjusting the inclination of the infant support surface.

The radiant heating support assembly includes the structural support, reflector arrangement, heating element and associated protective components.

Radiant Heating Assembly The radiant heating assembly is a safety- and performance-relevant subassembly of the BN-100 LUX.

The applicable manufacturer documentation identifies the radiant heating element, reflector arrangement and associated protective components. The manufacturer risk-management documentation identifies the air heater as a critical component.

Correct positioning, secure integration and appropriate protection of the heating assembly are relevant to the intended delivery of radiant thermal energy and to the control of risks associated with the heating function.

Temperature Monitoring and Control The temperature monitoring system incorporates the skin temperature sensor and associated electrical and electronic control components. The manufacturer risk-management documentation identifies the skin temperature sensor as a critical component.

Correct installation and connection of the sensor are relevant to the temperature monitoring and control functions. The applicable manufacturer documentation specifies characteristics associated with temperature control, sensor accuracy and alarm functions.

These characteristics are considered within the safety and performance requirements of the finished device.

Electrical and Electronic Assembly The electrical and electronic assembly includes components associated with power supply, electrical protection, control, temperature monitoring, heating and alarm functions.

The manufacturer risk-management documentation identifies the critical components described in Section 3.2.2.3 of this dossier. Their correct integration into the finished device is addressed within the manufacturer's production and quality-management framework.

Alarm and Safety Functions The BN-100 LUX incorporates alarm functions associated with specified abnormal conditions, including:

- temperature deviation;
- skin temperature sensor fault;
- over-temperature; and
- mains power failure.

The corresponding sensing, control and alarm components form part of the safety-related functional assemblies of the device. Their correct integration is relevant to the safety and intended operation of the finished device.

Intermediate Products and Subassemblies The principal intermediate assemblies identified from the available manufacturer documentation include, as applicable:

- mechanical base assembly;

- infant crib assembly;
- radiant heating assembly;
- operating and control assembly;
- electrical and electronic assemblies; and
- associated accessory assemblies.

These assemblies are progressively integrated during manufacture and final assembly of the device. The relevant manufacturing and assembly activities fall within the manufacturer's production and quality-management processes.

Production Controls The manufacturer's Risk Management Report identifies production and post-production activities as part of the overall risk-management process. These activities are to follow the manufacturer's applicable processes, methods and control measures.

The manufacturer documentation also identifies processes associated with product change control and customer feedback control. These processes form part of the documented framework for addressing manufacturing and post-production risks.

Detailed production inspection plans, work instructions, acceptance criteria, inspection frequencies, intermediate-product release records and manufacturing-stage verification records are not reproduced where they are not included in the supporting documentation available for this submission.

Documentary Basis This section is based primarily on the manufacturer's technical and risk-management documentation applicable to the **BN-100 LUX**.

The manufacturer's Risk Management Report, Document No. YF-7.3-24-02 (w), Version A/0, provides the principal documentary basis for identifying critical components, safety-related functions and manufacturing-process-related risks.

The manufacturer's user manual and other applicable technical documentation provide additional supporting information where relevant.

Validation and/or Evaluation of the Manufacturing Process

General The manufacture of the **BN-100 LUX** Infant Radiant Warmer involves mechanical processing, assembly of structural components, installation of electrical and electronic components, integration of the radiant heating system, installation of temperature-monitoring components, and final assembly of the device.

The available manufacturer documentation identifies manufacturing and post-production activities within the manufacturer's quality management and risk management framework.

These activities are considered in relation to the safety and performance of the finished device.

Manufacturing Process Evaluation The manufacturer's Risk Management Report considers risks associated with the manufacturing process and identifies production and post-production activities as part of the overall risk management process.

Manufacturing activities are required to follow the manufacturer's established processes, methods, and control measures. The manufacturer documentation also identifies processes associated with product change control and customer feedback control.

This documentation provides a basis for considering manufacturing- process-related risks within the manufacturer's quality management and risk management framework.

Process Validation The available documentation describes the principal manufacturing and assembly activities and their relationship with the manufacturer's quality management and risk management processes.

Detailed process-validation protocols, validation reports, predefined acceptance criteria, and individual validation records for each manufacturing operation have not been identified among the supporting documents currently available for this submission.

Accordingly, this section does not state specific validation results or numerical acceptance values that are not documented in the available manufacturer documentation.

Where a manufacturing process is subject to validation under the manufacturer's quality management system, the corresponding validation activities and records remain part of the controlled manufacturer documentation.

Where required for the final regulatory assessment, applicable process-validation documentation may be obtained from the manufacturer and incorporated into the supporting technical documentation.

Safety-Relevant Manufacturing Activities Particular attention is applicable to manufacturing and assembly activities associated with safety- and performance-relevant functions of the device, including:

- assembly of the radiant heating system;
- installation and positioning of the radiant heating element and reflector assembly;
- installation and connection of the skin temperature sensor;
- assembly of electrical and electronic control components;
- installation of power supply, protection, and switching components;
- installation of alarm-related components;
- assembly of protective structures and components; and
- final integration of the mechanical, electrical, and electronic subassemblies.

These activities are relevant to the safe and effective operation of the finished device and are considered within the manufacturer's production, quality management, and risk management processes.

Relationship with Risk Management Manufacturing-process-related risks are considered within the overall risk-control strategy described in the manufacturer's Risk Management Report. The report identifies **EN ISO 14971:2019** and **ISO/TR 24971:2020** as the applicable risk-management references.

The risk-management process provides a documented framework for the identification, evaluation, and control of risks associated with the device throughout its lifecycle.

Documentary Evidence The primary documentary basis for this section is the manufacturer's Risk Management Report, identified as follows:

- **Product:** BN-100 LUX Infant Radiant Warmer;
- **Document:** Risk Management Report;
- **Document No.:** YF-7.3-24-02 (w);
- **Version:** A/0;
- **Date:** 26 June 2023;
- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.

The manufacturer's user manual and other applicable technical documentation serve as supporting sources where relevant. The original manufacturer documentation is retained as supporting evidence in the regulatory dossier.

Any additional process-validation protocol, report, or record required for the final regulatory assessment may be obtained from the manufacturer and incorporated into the supporting documentation, where applicable.

Development of the Manufacturing Process

General The **BN-100 LUX** Infant Radiant Warmer is manufactured through mechanical processing, structural assembly, electrical and electronic assembly, integration of the radiant heating system, integration of the temperature-monitoring and control system, and final device assembly.

The available manufacturer documentation describes the principal construction and manufacturing elements of the device and places manufacturing activities within the manufacturer's quality-management and risk-management framework.

The manufacturing activities correspond to the principal functional assemblies and safety-related characteristics of the finished device.

Manufacturing Process Development The available manufacturer documentation describes the established manufacturing and assembly activities associated with the principal assemblies of the BN-100 LUX, including:

- processing and assembly of the principal mechanical structures;
- assembly of the infant crib and associated protective components;
- integration of the radiant heating assembly;
- integration of electrical and electronic components;
- installation of the temperature-monitoring system;
- integration of alarm and safety-related functions; and
- final assembly of the finished device.

A separate, detailed historical record of the successive development stages of the manufacturing process is not included in the documentation available for this submission.

Accordingly, this section describes the manufacturing process and its development only to the extent supported by the available manufacturer documentation.

Relationship Between Product Design and Manufacturing The manufacturing activities correspond to the principal functional assemblies of the BN-100 LUX, including:

- the base and mobile support structure;
- the infant crib and protective enclosure;
- the radiant heating assembly;
- the reflector and heating-element assembly;
- the operating and control assembly;
- the temperature-monitoring system;
- the electrical and electronic control system; and
- the alarm and safety-related components.

These assemblies are integrated during manufacturing and final assembly to form the finished Infant Radiant Warmer.

The relationship between product design, manufacturing activities, and the associated safety and performance characteristics is considered within the manufacturer's quality-management and risk-management processes.

The identification of critical components and safety-related functions in the manufacturer's risk-management documentation provides a documented link between product design and manufacturing controls.

Manufacturing Changes The manufacturer's Risk Management Report identifies product change control as part of the control of production and post-production activities.

Manufacturing and product changes are managed through the manufacturer's established change-control system.

The available documentation does not identify a specific manufacturing change, engineering change order, or individual process revision applicable to the device covered by this dossier.

Consequently, no specific manufacturing change or process revision is described in this section.

Transfer to Routine Production The available manufacturer documentation describes established manufacturing and assembly activities for the BN-100 LUX Infant Radiant Warmer.

The relationship between product design, manufacturing activities, quality management, and risk management is addressed within the manufacturer's documented processes.

Detailed design-transfer records, process-qualification packages, and production-readiness records are controlled manufacturer documents. They are not reproduced in this section where they have not been provided as supporting evidence for this submission.

Where required for the final regulatory assessment, applicable manufacturer documentation may be provided as additional supporting evidence.

Risk Management Considerations Manufacturing-process-related risks are considered within the manufacturer's Risk Management Report for the BN-100 LUX Infant Radiant Warmer.

The report identifies production and post-production activities as part of the overall risk-management process. Applicable manufacturing processes, methods, and control measures are to be followed in accordance with the manufacturer's established procedures, while product changes are managed through the manufacturer's change-control process.

The manufacturer's Risk Management Report identifies **EN ISO 14971:2019** and **ISO/TR 24971:2020** as the risk-management references applicable to the documented risk-management process.

This framework provides a documented basis for considering manufacturing-process-related risks throughout the device lifecycle.

Documentary Basis The information presented in this section is based on the manufacturer's technical and risk-management documentation applicable to the **BN-100 LUX** Infant Radiant Warmer.

The primary supporting document is identified as follows:

- **Document:** Risk Management Report;
- **Product:** BN-100 LUX Infant Radiant Warmer;
- **Document No.:** YF-7.3-24-02 (w);
- **Version:** A/0;
- **Date:** 26 June 2023;
- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.

The manufacturer's user manual and other applicable technical documentation serve as supporting sources where relevant. The original manufacturer documentation is retained as supporting evidence in the regulatory dossier.

This section does not introduce a manufacturing-process development history or specific development results beyond the information described in the available manufacturer documentation.

3.1.6 Control of Raw Material

Specifications

General The **BN-100 LUX** Infant Radiant Warmer incorporates mechanical, electrical, electronic, and patient-contact materials and components.

The manufacturer identifies specific materials and critical components relevant to the safety and performance of the device. The specifications presented in this section are based on the manufacturer's technical and risk-management documentation.

No additional material or component specification is introduced beyond the information documented by the manufacturer.

Patient-Contact Materials The manufacturer's Risk Management Report identifies the following materials as having direct contact with the infant:

- **Baby mattress:** cotton;
- **Skin temperature sensor probe:** stainless steel.

The manufacturer identifies skin contact exceeding 24 hours within a 30-day period for these materials.

These patient-contact materials are considered within the manufacturer's risk-management and biological-safety assessment. Their identification provides the material and patient-contact information used in the device risk assessment.

Critical Components The manufacturer identifies the following components as critical components of the **BN-100 LUX** Infant Radiant Warmer.

No.	Component	Model / Specification	Standard / Reference
1	Dual fuse input socket	SP-861	GB/T 17465.1-2022; GB/T 17465.6-2022
2	Power supply cord	3VTJ216A250V / 3VTJA10A2m	DIN VDE 0620-1 (VDE 0620-1):2010-02
3	Auxiliary socket outlets	ST-A02	EN 60320-1:2015
4	Relay	HF115F012-1HS3	GB/T 21711.1-2008
5	Rocker switch	KCD2	ANSI/UL 61058-1
6	Fuse	F5AL250V	EN 60127-1:2006+A1:2011+A2:2015
7	Rechargeable battery	B80H7A2H	As identified in the manufacturer's documentation
8	Optical coupler	MOC3083	DIN EN 60747-5
9	Printed circuit board	NPG-180BH	GB/T 4725-1992; GB/T 6040-2019; GB/T 19466.1-2004; GB/T 19466.2-2004; GB/T 19466.3-2004; ISO 11358-1:2014
10	Internal wire	0.5 / 1.0 / 2.5 / 4.0 mm^2	GB/T 5023.5-2008 / IEC 60227-5:2003; JB/T 8734.3-2016
11	Skin temperature sensor	T10-AS	MDD Annex II, excluding Section 4
12	Air heater	EQ-076-A	IEC 60601-1
13	Thyristors	BTA12-600B	IEC 60601-1
14	Audible alarm	SFM-27	IEC 60601-1
15	Switching power supply	RPS-45-12	BS EN 60601-1:2006+A12:2014
16	Filter	RV410-6	EN 60939-1:2010; EN 60939-2:2005

Component Identification and Control The critical components listed above are addressed within the manufacturer's product-design, manufacturing, and risk-management activities.

Where applicable, the manufacturer's documentation identifies technical standards and associated certification or registration references for individual components.

The component models, technical references, and associated documentation provide a basis for component identification and control within the manufacturer's production system.

Detailed supplier specifications, incoming-inspection requirements, supplier-qualification criteria, and acceptance criteria are manufacturer-controlled documentation and are not reproduced in this section where they have not been provided as supporting evidence for this submission.

Relationship with Risk Management The identification of critical materials and components forms part of the manufacturer's risk-management process. Particular consideration is given to components associated with:

- patient contact;
- radiant heating;
- temperature measurement and control;
- electrical power and protection;
- electronic control;
- alarm functions; and
- other safety-related functions of the device.

The manufacturer's Risk Management Report provides the primary documentary basis for the identification of the materials and critical components described in this section.

The component specifications and associated technical references are considered as part of the documented basis for controlling components relevant to device safety and performance.

Documentary Basis The information presented in this section is based primarily on the manufacturer's Risk Management Report for the **BN-100 LUX** Infant Radiant Warmer, identified as follows:

- **Document:** Risk Management Report;
- **Document No.:** YF-7.3-24-02 (w);
- **Version:** A/0;
- **Date:** 26 June 2023;
- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.

The manufacturer's technical documentation and user manual serve as supporting sources where relevant. The original manufacturer documentation is retained as supporting evidence in the regulatory dossier.

Justification of Specifications

General The specifications identified for the materials and components used in the **BN-100 LUX** Infant Radiant Warmer are considered in relation to the intended function, safety, and performance characteristics of the device.

For a neonatal radiant warmer, relevant specifications concern the mechanical structure, radiant heating system, temperature-monitoring and control system, electrical power and protection, alarm functions, and patient-contact materials.

The manufacturer's documentation identifies the principal materials and critical components considered relevant to device safety and performance. Their specifications are therefore considered according to the functions they perform within the finished device.

Patient-Contact Materials The manufacturer's Risk Management Report identifies the following materials as having direct contact with the infant:

- **Baby mattress:** cotton;
- **Skin temperature sensor probe:** stainless steel.

The manufacturer identifies skin contact exceeding 24 hours within a 30-day period for these materials.

The mattress provides the infant support surface, while the skin temperature sensor provides the interface required for measuring the infant's temperature.

These materials are relevant to the intended patient-contact conditions and are considered within the manufacturer's risk-management and biological-safety assessment.

Critical Components The manufacturer identifies critical electrical and electronic components used in the **BN-100 LUX**. Their functions include:

- electrical power input and protection;
- power supply and switching;
- electrical isolation and control;
- temperature sensing and monitoring;
- radiant heating;
- audible alarm functions;
- printed circuit boards and internal wiring; and
- other safety-related electrical functions.

The component models and applicable technical references, where documented, provide a basis for identifying the components and relating them to their intended functions within the device.

The manufacturer's Risk Management Report provides the primary documentary basis for considering these components within the device risk-management process.

Relationship with Device Safety and Performance The specifications of materials and components are relevant to the safety and performance characteristics of the **BN-100 LUX**, particularly the following functions:

- **Radiant heating:** the heating components are relevant to the generation and delivery of thermal energy;
- **Temperature monitoring and control:** the sensing components are relevant to monitoring the infant's thermal environment;
- **Electrical power and protection:** the power-supply, switching, and protection components are relevant to the electrical system;
- **Alarm functions:** the alarm-related components are relevant to indicating defined abnormal or safety-related conditions; and
- **Mechanical structure and patient contact:** the structural and patient-contact materials are relevant to supporting the infant and providing the intended interface between the device and the patient.

The manufacturer's risk-management documentation considers hazards and risk-control measures associated with device construction, materials, components, and manufacturing activities.

The identified specifications therefore contribute to the documented technical basis for controlling materials and components relevant to device safety and performance.

Scope of the Documented Specifications The available manufacturer documentation identifies principal materials and critical components and provides applicable technical references where documented.

Detailed supplier specifications, complete material certificates, individual component datasheets, and detailed acceptance criteria are not reproduced in this section where they are not included in the supporting documentation available for this submission.

No independent acceptance criteria, analytical limits, or material-performance requirements are introduced beyond those documented by the manufacturer.

Where additional detail is required for the regulatory assessment, the corresponding manufacturer documentation may be provided as supporting evidence.

Regulatory Conclusion The principal materials and critical components identified for the **BN-100 LUX** are considered in relation to their intended functions, patient-contact characteristics, and relevance to device safety and performance.

The manufacturer's Risk Management Report provides the primary documentary basis for identifying these materials and components and considering their associated risks.

Together with the applicable component references, this information supports the documented technical rationale for the identification and control of safety- and performance-relevant materials and components used in the device.

Documentary Basis The information presented in this section is based primarily on the manufacturer's Risk Management Report, identified as follows:

- **Product:** BN-100 LUX Infant Radiant Warmer;

- **Document:** Risk Management Report;
- **Document No.:** YF-7.3-24-02 (w);
- **Version:** A/0;
- **Date:** 26 June 2023;
- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.

The manufacturer's technical documentation and user manual serve as supporting sources where relevant. The original manufacturer documentation is retained as supporting evidence in the regulatory dossier.

3.2 Finished Product

3.2.1 Description and Composition of the Finished Product

General Description

The finished product covered by this application is the **BN-100 LUX** neonatal radiant warmer.

The device is an active electrical medical device intended to provide controlled radiant thermal support to newborns and neonates requiring a controlled warming environment.

It provides non-contact radiant heat through an overhead heating assembly and incorporates a temperature monitoring and control system using a skin temperature sensor.

The finished device consists of a mechanical support structure, infant bed, radiant heating assembly, control and operating system, temperature monitoring system, electrical and electronic components, alarm functions and associated accessories.

The description presented here is based on the manufacturer documentation applicable to the BN-100 LUX.

Device Configuration

The principal assemblies of the finished BN-100 LUX include:

- a base and supporting structure;
- an infant bed with protective side baffles;
- an overhead radiant heating assembly;
- a radiant reflector assembly;
- a heating tube;
- a control and operating system;
- a skin temperature sensing system;
- electrical and electronic control components;
- audible alarm functions;

- an intravenous support pole or hook;
- an X-ray cassette tray; and
- associated mechanical and electrical components.

Depending on the supplied configuration, additional accessories or functional modules may be provided. The applicable configuration is determined by the relevant manufacturer configuration and packing documentation.

Principal Materials

The available manufacturer documentation identifies the following materials for the specified components or associated equipment:

- **Carbon steel:** radiant heating tube;
- **Lucite/Plexiglas:** protective side baffles; and
- **Polycarbonate:** suction bottle.

The available documentation does not provide a complete material declaration covering every component of the finished device. Exact material grades, chemical compositions, coatings, adhesives and supplier-level specifications are not assigned where they are not explicitly documented.

Temperature Monitoring and Control

The BN-100 LUX incorporates a skin temperature monitoring system. The sensor provides temperature information to the control system for thermal monitoring and control.

The manufacturer documentation specifies the following relevant temperature-control characteristics:

- skin temperature control range: 32–38 °C;
- temperature control precision: ≤ 0.5 °C;
- skin temperature sensor precision: ≤ 0.3 °C;
- overheating level 1: 39 °C;
- overheating level 2: 40 °C; and
- temperature deviation alarm: ± 1 °C.

The skin temperature sensor shall be positioned and used in accordance with the manufacturer's instructions.

Heating System

Thermal support is provided by the overhead radiant heating system.

The manufacturer documentation identifies an explosion-proof infrared heating tube rated at **800 W**.

The heating source is positioned above the infant bed and transfers heat through non-contact radiant heating. A reflector assembly directs radiant energy toward the infant bed area.

The heating system incorporates safety functions associated with temperature monitoring and overheating protection.

Alarm and Safety Functions

The finished device incorporates alarm and safety functions associated with temperature monitoring, heating control and electrical operation.

The manufacturer documentation identifies alarm functions associated with:

- mains power failure;
- skin temperature sensor failure;
- overheating; and
- temperature deviation.

These functions are intended to alert the operator to specified abnormal conditions. Their detailed characteristics and the corresponding operator responses are described in the manufacturer's instructions for use and applicable technical documentation.

Electrical and Electronic Components

The finished device contains electrical and electronic components required for power supply, heating control, temperature monitoring and alarm functions.

The manufacturer documentation identifies, among other components:

- power supply and protection components;
- switching and relay components;
- fuses;
- a rechargeable battery associated with the alarm function;
- a printed circuit board and internal wiring;
- a skin temperature sensor;
- a heating element;
- an audible alarm; and
- electronic switching and control components.

Detailed component specifications are addressed in the corresponding technical sections of the dossier, where applicable.

Patient Contact

The manufacturer risk-management documentation identifies the infant mattress and skin temperature sensor probe as components having patient contact.

The available risk-management documentation identifies patient contact occurring for more than 24 hours within a 30-day period.

The applicable patient-contact materials and associated biological safety considerations are addressed in the relevant material and biological evaluation documentation.

Accessories and Optional Components

Depending on the configuration, the BN-100 LUX may be supplied with or equipped with manufacturer-identified accessories and auxiliary functions, including:

- skin temperature sensor;
- X-ray cassette tray;
- intravenous support;
- spotlight;
- oxygen cylinder support;
- LED phototherapy unit; and
- low-pressure suction equipment.

Optional components are considered part of the submitted configuration only when specified for the applicable device configuration.

Non-Sterile and Reusable Device

The BN-100 LUX is a reusable medical device and is not presented as a sterile, single-use medical device.

Cleaning and disinfection shall be performed in accordance with the manufacturer's instructions for use and applicable institutional procedures.

3.2.2 Manufacturing

Manufacturer(s)

Manufacturer Identification The manufacturer identified for the finished **BN-100 LUX** Neonatal Radiant Warmer is:

- **Manufacturer:** Zhengzhou Dison Instrument and Meter Co., Ltd.;
- **Address:** No. 67, Jinsuo Road, High-Tech Zone Development, Zhengzhou, Henan, China;
- **Postal Code:** 450001.

Manufacturing Site The available manufacturer documentation identifies the address above as the manufacturer's address. It does not establish whether additional manufacturing facilities or subcontracted manufacturing sites are involved.

No additional manufacturing site or subcontractor is identified in this section in the absence of supporting documentation.

Quality Management Documentation The manufacturer's quality management system certification information is presented in the manufacturer's documentation section earlier in this dossier. Certificate details are not repeated here.

Composition

General The finished **BN-100 LUX** Neonatal Radiant Warmer is an active electrical medical device composed of mechanical, electrical and electronic assemblies integrated into a single system to provide controlled radiant thermal support to newborns and neonates.

Its principal assemblies include the structural components, radiant heating system, temperature monitoring and control components, electrical power and protection components, alarm components and associated accessories.

The composition described below is based on the applicable manufacturer documentation, including the manufacturer's risk-management documentation.

Mechanical Assemblies The principal mechanical assemblies of the finished device include:

- the base and mobile support structure;
- the infant crib;
- the protective side panels;
- the radiant heating support structure;
- the reflector support;
- the operating assembly;
- accessory support components; and
- the X-ray cassette support and tray arrangement.

According to the manufacturer's risk-management documentation, the base incorporates aluminium alloy extrusions and processed, welded and surface-finished structural components.

The crib incorporates formed and welded steel components with protective surface finishing. It also includes an X-ray cassette window or tray arrangement and a mechanism for adjusting the crib inclination.

Radiant Heating Assembly The radiant heating assembly is a principal functional assembly of the finished device. According to the applicable manufacturer documentation, it includes:

- an aluminium alloy support structure;
- a stainless steel reflector;
- a fixed explosion-proof quartz infrared heating tube;
- a protective burn net; and
- an adjustable-angle LED white light.

The assembly is positioned above the infant crib and provides the radiant thermal energy required for the intended thermal-support function.

Detailed operating characteristics of the heating system are addressed in the relevant technical sections of the dossier.

Control and Operating Assembly The operating assembly contains components required for operation, monitoring and control of the device.

The manufacturer's documentation identifies power transformers and control circuit boards installed within this assembly.

The assembly also provides support for certain accessories, including the infusion hook and carrier tray, depending on the applicable configuration.

Temperature Monitoring System The finished device incorporates a skin temperature monitoring system. The principal patient-contact component of this system is the skin temperature sensor probe, identified in the applicable manufacturer documentation as model **T10-AS**.

The sensor provides temperature information to the control system for monitoring and regulation of the infant's thermal environment.

The manufacturer's risk-management documentation identifies the skin temperature sensor as a critical component of the device.

Electrical and Electronic Composition The electrical and electronic composition of the finished device includes components associated with power input, electrical protection, switching, control, temperature sensing, heating and alarm functions.

The manufacturer's risk-management documentation identifies the following critical components:

- dual-fuse input socket;
- power supply cord;
- auxiliary socket outlets;
- relay;
- rocker switch;
- fuse;
- rechargeable battery;

- optocoupler;
- printed circuit board;
- internal wiring;
- skin temperature sensor;
- air heater;
- thyristors;
- audible alarm; and
- switching power supply.

These components are identified as relevant to the device's functionality and safety within the manufacturer's risk-management process. Their detailed specifications and applicable references are addressed in the relevant component and technical sections of the dossier.

Patient-Contact Components The manufacturer's risk-management documentation identifies the following principal patient-contact components and materials:

- **Baby mattress:** cotton;
- **Skin temperature sensor probe:** stainless steel.

The manufacturer identifies direct patient contact exceeding 24 hours within a 30-day period for these components.

These components and their identified materials are included in the manufacturer's safety and risk-management assessment.

Accessories and Configurable Components Depending on the applicable configuration, the finished **BN-100 LUX** may be supplied with manufacturer-identified accessories and optional components, including:

- skin temperature sensor;
- X-ray cassette tray;
- infusion hook or support;
- spotlight;
- oxygen support components;
- phototherapy equipment; and
- suction equipment.

Optional components form part of the submitted finished-device configuration only when specified for that configuration. The applicable configuration and supplied accessories are to be identified from the manufacturer's relevant configuration and packing documentation.

Composition and Risk Management The identification of critical components, patient-contact components and safety-related assemblies forms part of the manufacturer's risk-management process for the intended use of the **BN-100 LUX**.

The composition presented in this section is based on the applicable manufacturer documentation. Detailed component specifications and risk assessments are addressed in their respective sections of the dossier.

Description of the Manufacturing Process and Operational Controls

General The **BN-100 LUX** Neonatal Radiant Warmer is manufactured and assembled by Zhengzhou Dison Instrument and Meter Co., Ltd.

The available manufacturer documentation describes the principal assemblies, materials, critical components and safety-related characteristics of the finished device.

The documented manufacturing and assembly activities involve the mechanical structure, radiant heating system, temperature monitoring and control system, electrical and electronic assemblies, and associated accessories.

The available documentation does not provide a complete manufacturing sequence, detailed work instructions, process parameters, acceptance criteria or complete production records. Accordingly, this section describes the documented activities without introducing additional manufacturing operations or undocumented process requirements.

Mechanical Structure and Base Assembly According to the manufacturer documentation, the base structure incorporates aluminium alloy extrusions. The documented processing activities for structural components include processing, welding and surface spraying.

The lower part of the device may be equipped with four mobile castors with brakes, depending on the applicable configuration.

The crib incorporates steel components manufactured using forming, bending and welding operations, followed by surface finishing. It also incorporates the documented X-ray cassette arrangement and a mechanism for adjusting the crib inclination. Protective Plexiglas panels are incorporated into the crib assembly.

These components are integrated to provide the mechanical support for the infant crib, operating section and associated assemblies.

The specific mechanical inspection procedures and acceptance limits are not detailed in the available documentation.

Radiant Heating Assembly According to the applicable manufacturer documentation, the radiant heating assembly comprises:

- an aluminium alloy structural assembly;
- a stainless steel reflector;
- a fixed explosion-proof quartz infrared heating tube; and
- a protective net in front of the heating element.

The device also incorporates an adjustable-angle LED white light.

The radiant heating assembly is integrated with the control system to provide the intended controlled thermal environment for the patient. The heating element and its associated components are identified as safety-relevant components in the manufacturer documentation and are considered within the manufacturer's quality and risk-management framework.

Operating and Control Assembly The operating section incorporates processed aluminium alloy components and houses the electrical and electronic assemblies required for operation of the device.

The manufacturer documentation identifies power transformers, control circuit boards and associated electrical components within this assembly.

The operating section is fitted with its corresponding cover and may support additional components, such as the infusion hook and carrier tray, depending on the applicable configuration.

The assembly integrates the mechanical, electrical and electronic components into the finished device. The detailed assembly sequence and individual production instructions are not provided in the available documentation.

Temperature Monitoring and Control The **BN-100 LUX** incorporates a skin temperature sensor and a temperature control system. The documented skin temperature sensor is identified as model **T10-AS**.

The sensor monitors the infant's skin temperature and provides temperature information to the control system.

The manufacturer's risk-management documentation identifies the following characteristics of the finished device:

- control temperature range: 32–38 °C;
- display temperature range: 10–45 °C;
- control precision: ≤ 0.5 °C;
- skin temperature sensor precision: ≤ 0.3 °C;
- over-temperature protection at approximately 40 °C;
- temperature deviation alarm at approximately ±1 °C; and
- alarm in the event of a skin temperature sensor short circuit or open circuit.

These values describe the documented operating and safety characteristics of the finished device. They are not presented here as individual manufacturing-process acceptance criteria.

Electrical and Electronic Assembly The electrical and electronic assembly incorporates components required for power supply, switching, control, protection, temperature monitoring, heating and alarm functions.

The manufacturer's risk-management documentation identifies the following critical components:

- dual-fuse input socket;
- power supply cord;
- auxiliary socket outlets;
- relay;
- rocker switch;

- fuse;
- rechargeable battery;
- optocoupler;
- printed circuit board;
- internal wiring;
- skin temperature sensor;
- air heater;
- thyristors;
- audible alarm;
- switching power supply; and
- electrical filter.

These components are identified in the manufacturer's risk-management documentation as relevant to the device's functionality and safety.

Detailed component-level production specifications, supplier requirements and individual acceptance criteria are not provided in the documentation available for this section.

Alarm and Safety-Related Operations The finished **BN-100 LUX** incorporates safety-related alarm functions documented by the manufacturer, including:

- over-temperature alarm and heating cut-off;
- temperature deviation alarm;
- skin temperature sensor short-circuit or open-circuit alarm;
- power failure alarm; and
- manual-mode time alarm.

These functions are identified as safety-related characteristics of the device in the manufacturer's risk-management documentation.

The corresponding electrical and electronic components are integrated during assembly. The specific production test procedures and acceptance criteria for each alarm function are not detailed in the documentation available for this section.

Integration of the Finished Device Final assembly integrates the mechanical structure, infant crib, radiant heating assembly, operating and control assembly, temperature monitoring system, electrical and electronic components, alarm functions and applicable accessories.

The available documentation does not provide a separate detailed production flow chart or a complete set of manufacturing work instructions.

The finished-device assembly is considered within the manufacturer's applicable quality-management and risk-management processes, including production and post-production activities.

Manufacturing Controls and Risk Management Manufacturing-related risks are addressed within the manufacturer's risk-management documentation applicable to the **BN-100 LUX**.

The documentation identifies manufacturing-process risk assessment as part of the overall risk-management activities. It also addresses production and post-production activities, product change control and customer feedback control.

Manufacturing and assembly activities are therefore considered within the manufacturer's quality-management and risk-management framework.

No additional manufacturing procedure, process parameter, acceptance criterion or inspection record is specified here unless supported by the available manufacturer documentation.

Controls of Critical Steps and Intermediate Steps

General The manufacturer documentation available for the **BN-100 LUX** identifies components, assemblies and functions relevant to the safety and performance of the finished device.

The applicable risk-management documentation also addresses manufacturing-process-related risks.

The documentation available for this submission does not provide a separate detailed set of manufacturing-stage acceptance criteria, intermediate-product specifications or inspection records for each individual production step.

Accordingly, this section describes the documented critical components, assemblies and safety-related functions without introducing unsupported manufacturing controls or acceptance criteria.

Mechanical Assemblies The manufacturer documentation identifies the following principal mechanical assemblies:

- the base structure incorporating aluminium alloy extrusions;
- the infant crib incorporating steel components;
- the crib inclination mechanism;
- the Plexiglas protective panels;
- mobile castors, where provided; and
- associated supporting structures.

The documented manufacturing operations include processing, welding and surface spraying of relevant structural components.

These operations are considered within the manufacturer's manufacturing and quality-management framework. However, separate intermediate-product acceptance criteria and inspection limits for the individual mechanical manufacturing operations are not detailed in the available documentation.

Radiant Heating Assembly The radiant heating assembly incorporates components identified in the manufacturer documentation, including:

- aluminium alloy structural components;
- stainless steel reflector;

- fixed explosion-proof quartz infrared heating tube;
- protective net; and
- adjustable-angle LED white light.

The applicable risk-management documentation identifies the air heater as a critical component.

The heating assembly is therefore considered a safety-relevant assembly within the manufacturer's production, quality-management and risk-management processes.

A separate detailed intermediate inspection procedure or acceptance record specifically applicable to this assembly is not provided in the documentation available for this submission.

Temperature Monitoring System The **BN-100 LUX** incorporates a skin temperature sensor identified by the manufacturer as model **T10-AS**.

The applicable risk-management documentation identifies the sensor as a critical component. Documented characteristics relevant to temperature monitoring include:

- skin temperature sensor precision: $\leq 0.3\,^{\circ}\mathrm{C}$;
- sensor short-circuit alarm; and
- sensor open-circuit alarm.

The sensor is therefore considered a safety-related component of the temperature monitoring system.

A separate production-stage inspection procedure or acceptance record specifically applicable to this sensor is not provided in the available documentation.

Electrical and Electronic Components The applicable risk-management documentation identifies the following critical components:

- dual-fuse input socket;
- power supply cord;
- auxiliary socket outlets;
- relay;
- rocker switch;
- fuse;
- rechargeable battery;
- optocoupler;
- printed circuit board;
- internal wiring;
- skin temperature sensor;

- air heater;
- thyristors;
- audible alarm;
- switching power supply; and
- electrical filter.

The manufacturer documentation associates these components with their respective component references and, where applicable, technical or certification information. Their identification supports consideration within the manufacturer's production, assembly and risk-control framework.

Separate production inspection records and individual acceptance criteria for the integration of each component are not provided in the documentation available for this submission.

Safety-Related Functions The applicable risk-management documentation identifies the following safety-related characteristics of the **BN-100 LUX**:

- over-temperature protection with heating cut-off and audible and/or visual alarm;
- temperature deviation alarm;
- skin temperature sensor short-circuit and open-circuit alarms;
- power failure alarm; and
- manual-mode alarm.

These characteristics form part of the manufacturer's documented risk-control framework.

The available documentation does not provide a complete production-stage verification procedure for each safety-related function. No additional verification method, acceptance criterion or production record is therefore specified in this section unless supported by the applicable manufacturer documentation.

Manufacturing Process Risk Control The manufacturer's applicable risk-management documentation includes manufacturing-process risk assessment within the overall risk-management activities.

It also states that production and post-production activities are to follow the applicable processes, methods and measures established by the manufacturer, including, where applicable, product change-control and customer-feedback-control processes.

This documentation provides the basis for considering manufacturing-process-related risks associated with the finished device.

Intermediate Products and Assemblies The manufacturer documentation describes the principal components and assemblies of the **BN-100 LUX**, but does not provide a separate formal classification of intermediate products or a complete list of intermediate-product specifications.

Accordingly, no additional intermediate products, manufacturing-stage specifications or acceptance criteria are introduced in this section.

Process Validation and/or Evaluation

General The manufacture of the **BN-100 LUX** Neonatal Radiant Warmer involves mechanical fabrication, assembly of the radiant heating system, integration of electrical and electronic components, and integration of the temperature monitoring and control system.

The available manufacturer documentation describes principal manufacturing operations, critical components and manufacturing-process-related risks.

A separate comprehensive process-validation report or complete process-qualification dossier specific to the **BN-100 LUX** has not been identified among the documentation available for this submission.

Manufacturing Processes Addressed in the Available Documentation The manufacturer documentation identifies manufacturing and assembly operations associated with the principal device assemblies, including:

- processing, welding and surface spraying of the aluminium alloy base structure;
- bending, welding and surface spraying of the steel crib;
- fabrication and assembly of the radiant heating structure;
- integration of the infrared heating element and reflector;
- assembly and integration of electrical and electronic components; and
- integration of the skin temperature sensor and temperature control system.

These operations are described at a general technical level. Separate process-validation protocols, predefined validation acceptance criteria and documented validation results for each individual operation have not been identified in the available documentation.

Risk-Based Consideration The applicable manufacturer risk-management documentation includes manufacturing-process risk assessment within the overall risk-management activities.

It identifies safety-related components and functions and considers risks associated with the manufacture and operation of the device.

The manufacturer documentation also addresses production and post-production activities, including applicable product change-control and customer-feedback-control processes.

These elements provide the available documentary basis for considering manufacturing-process-related risks. However, risk-management documentation alone does not establish that each manufacturing process has undergone formal validation.

Process Validation Evidence The following specific process-validation evidence has not been identified in the documentation available for this submission:

- dedicated process-validation protocols;
- documented validation acceptance criteria for individual manufacturing processes;
- process-qualification reports;
- validation records for welding, surface spraying or other mechanical fabrication processes;

- documented revalidation criteria or revalidation reports;
- complete validation records for production equipment or manufacturing lines.

No validation result or acceptance criterion is stated in this section unless explicitly supported by the available manufacturer documentation.

Process Evaluation The available information permits a documentary-level evaluation of the manufacturing process through:

- identification of the principal manufacturing and assembly operations;
- identification of critical components and safety-related assemblies;
- consideration of manufacturing-process-related risks; and
- reference to the manufacturer's quality-management and risk-management framework.

This evaluation describes the information available for regulatory review and does not replace formal process validation or qualification where required for a specific manufacturing process.

Applicability of Process Validation The need for formal process validation depends on the nature of the manufacturing process and whether its output can be fully verified by subsequent inspection or testing.

The available documentation does not identify a separate list of processes designated by the manufacturer as requiring special process validation.

Accordingly, individual manufacturing operations are not classified in this section as validated or non-validated in the absence of corresponding manufacturer evidence.

Conclusion The available manufacturer documentation identifies principal manufacturing and assembly operations and addresses manufacturing-process-related risks within the manufacturer's risk-management framework.

Nevertheless, a separate comprehensive process-validation dossier specific to the **BN-100 LUX** has not been identified among the documents available for this submission.

No unsupported statement of successful process validation is therefore made in this section.

3.2.3 Control of the Finished Product

General

The **BN-100 LUX** Infant Radiant Warmer is a finished medical device intended to provide a controlled thermal environment for newborn and infant patients requiring thermal support.

The manufacturer documentation identifies safety-related characteristics, critical components and functions relevant to the finished device. This section presents these characteristics in relation to the documented risk-control measures and the available finished-product documentation.

Safety-Related Finished Product Functions

The manufacturer's Risk Management Report identifies the following safety-related functions and alarm conditions:

- Over-temperature protection with heating cut-off and audible and visual alarm;
- Temperature deviation alarm;
- Skin temperature sensor short-circuit alarm;
- Skin temperature sensor open-circuit alarm;
- Power failure alarm;
- Manual-mode alarm.

These functions form part of the documented safety and risk-control framework for the finished device.

Critical Components

The manufacturer's Risk Management Report identifies critical components associated with the electrical, electronic, heating, temperature-monitoring and alarm functions of the device, including:

- Dual-fuse input socket;
- Power supply cord;
- Auxiliary socket outlets;
- Relay;
- Rocker switch;
- Fuse;
- Rechargeable battery;
- Optocoupler;
- Printed circuit board;
- Internal wiring;
- Skin temperature sensor;
- Air heater;
- Thyristors;
- Audible alarm;
- Switching power supply;
- Electrical filter.

These components are relevant to the documented safety-related functions and characteristics of the finished device.

Patient-Contact Materials

The manufacturer identifies the following components as involving direct patient contact:

- Cotton baby mattress;
- Stainless-steel skin temperature sensor probe.

The manufacturer's risk-management documentation considers these patient-contact components in relation to the associated biological safety considerations.

Finished Product Safety Considerations

The finished device incorporates mechanical, thermal, electrical, electronic and temperature-monitoring functions.

The manufacturer's risk-management documentation addresses risks associated with these functions and identifies corresponding risk-control measures.

The characteristics and functions described in this section complement the specifications presented in Section 3.3.4.1 and their justification in Section 3.3.4.5.

Documentation Limitation

The documentation available for this submission does not include a complete, separate finished-product inspection protocol specifying individual acceptance criteria and inspection records for every manufactured unit.

Accordingly, this section does not claim the existence of unit-specific inspection results, batch-release records or acceptance criteria that are not described in the available documentation.

The information presented is limited to the finished-product characteristics, critical components and functions documented by the manufacturer.

Documentary Basis

This section is based primarily on the following manufacturer documentation:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

Specifications

General The specifications of the finished **BN-100 LUX** Infant Radiant Warmer are based on the applicable manufacturer documentation available for the device.

They cover the documented electrical, environmental, thermal, alarm, storage and mechanical characteristics.

Electrical Specifications The manufacturer documentation specifies the following electrical characteristics:

- Rated supply voltage: 220 ± 22 V;
- Rated frequency: 50 ± 1 Hz;
- Rated power: 1000 VA;
- Heater power: 800 W.

Environmental Operating Conditions The documented operating conditions are:

- Ambient temperature: 18–30 °C;
- Relative humidity: 30–75 %;
- Atmospheric pressure: 700–1060 hPa;
- Air velocity: less than 0.3 m/s.

Temperature Control Specifications The manufacturer documentation specifies the following temperature-related characteristics:

- Temperature display range: 10–45 °C;
- Control temperature range: 32–38 °C;
- Temperature control precision: ≤ 0.5 °C;
- Bed temperature uniformity: ≤ 2 °C;
- Skin temperature sensor precision: ≤ 0.3 °C.

The device incorporates a skin temperature monitoring system for monitoring the newborn's temperature during operation.

Alarm and Protection Specifications The manufacturer documentation identifies the following alarm and protection characteristics:

- Over-temperature protection with heating cut-off and audible and visual alarm;
- First over-temperature alarm threshold: 39 °C;
- Second over-temperature alarm threshold: 40 °C;
- Temperature deviation alarm: ±1 °C;
- Skin temperature sensor short-circuit alarm;
- Skin temperature sensor open-circuit alarm;
- Power failure alarm; and
- Manual-mode alarm after 15 minutes.

The documented power failure alarm duration is approximately 10 minutes.

The alarm sound level is specified as not exceeding 80 dB, with a sound level of at least 65 dB at a distance of 3 m.

Storage Conditions The manufacturer documentation specifies the following storage conditions:

- Storage temperature: −40– + 55 °C;
- Relative humidity: not more than 95 %;
- Atmospheric pressure: 500–1060 hPa.

Mechanical Specifications The manufacturer documentation specifies the following characteristics for the patient support structure:

- Crib inclination: up to 10°;
- Three-step inclination adjustment;
- Mattress maximum load: 100 N (10 kg);
- Tray maximum load: 20 N (2 kg).

Documentary Basis The specifications presented in this section are based on the manufacturer documentation available for the **BN-100 LUX** Infant Radiant Warmer, including:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

The specifications above are limited to the characteristics identified in the cited manufacturer documentation.

Justification of Specifications

General The specifications of the **BN-100 LUX** Infant Radiant Warmer are based on the applicable manufacturer documentation and relate to the intended function, documented performance characteristics and safety-related functions of the device.

The justification presented in this section explains the relationship between the specifications in Section 3.3.4.1, the intended function of the device and the manufacturer's risk-management documentation.

Electrical Specifications The rated supply voltage, frequency, rated power and heater power define the electrical operating conditions specified by the manufacturer.

These characteristics describe the device's electrical configuration and provide the basis for connection to the specified electrical supply.

The manufacturer's Risk Management Report identifies electrical components relevant to device safety, including the input socket, power supply cord, fuse, relay, rocker switch, switching power supply and filter.

Environmental Specifications The specified operating temperature, relative humidity, atmospheric pressure and air velocity define the environmental conditions documented for operation of the **BN-100 LUX**.

The specified storage temperature, relative humidity and atmospheric pressure define the documented environmental conditions applicable during storage.

These limits are included in the specifications to describe the operating and storage conditions established by the manufacturer.

Temperature Control Specifications The temperature display range, control temperature range and temperature control precision are directly related to the principal function of the **BN-100 LUX**: providing controlled radiant heat to newborns and infants requiring a regulated thermal environment.

The specified bed temperature uniformity relates to heat distribution over the patient support area.

The skin temperature sensor precision relates to monitoring the patient's skin temperature and to the operation of the temperature control system.

These characteristics are therefore relevant to the intended thermal function and temperature monitoring capabilities of the device.

Alarm and Protection Specifications The documented alarm and protection characteristics are related to safety-related functions identified in the manufacturer's Risk Management Report.

The over-temperature protection, heating cut-off and alarm functions address excessive temperature conditions.

The documented over-temperature thresholds provide indications associated with the specified temperature limits. The temperature deviation alarm indicates deviation from the applicable temperature condition.

The skin temperature sensor short-circuit and open-circuit alarms address faults affecting the temperature monitoring system.

The power failure and manual-mode alarms provide additional safety-related indications identified by the manufacturer.

These characteristics are included in the specifications because of their relationship to the device's documented safety functions and risk-control measures.

Mechanical Specifications The documented crib inclination, mattress load and tray load specifications define the applicable mechanical limits of the patient support structure and associated components.

The crib inclination specification relates to the positioning function of the patient support structure. The mattress and tray load specifications define the maximum loads stated by the manufacturer for these components.

No additional mechanical limits or acceptance criteria are introduced in this section.

Patient-Contact Materials The manufacturer identifies the cotton baby mattress and stainless-steel skin temperature sensor probe as components involving direct patient contact.

These components are considered within the manufacturer's risk-management documentation. Their identification in the finished-product documentation is therefore linked to the manufacturer's documented assessment of patient-contact-related risks.

Risk Management Relationship The specifications of the finished product are considered alongside the safety-related components and functions identified in the manufacturer's Risk Management Report.

The documented electrical, environmental, thermal, mechanical and alarm characteristics describe relevant aspects of the **BN-100 LUX** and their relationship to its intended function and safety-related performance.

The justification in this section is limited to the relationship supported by the available manufacturer documentation.

Limitation of the Justification The available manufacturer documentation does not include a separate document entitled "Justification of Finished Product Specifications".

Accordingly, this section provides a structured explanation of the relationship between the documented specifications, the intended function of the device and the available risk-management information. It does not introduce independent performance claims or acceptance criteria.

Documentary Basis This section is based primarily on the following manufacturer documentation:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.2.4 Reference Standards

General

The **BN-100 LUX** Infant Radiant Warmer is designed and manufactured with reference to applicable technical and safety requirements identified in the manufacturer documentation available for the device.

The standards and reference documents identified in the available documentation are presented below according to their documented scope.

No additional standard or conformity claim is introduced where the corresponding manufacturer documentation is not available in the submission file.

IEC 60601-1

IEC 60601-1 is identified as the applicable general standard for the basic safety and essential performance of medical electrical equipment.

The available manufacturer documentation indicates that compliance with the applicable requirements of IEC 60601-1 is relevant to the electrical safety and essential performance of the device.

The corresponding test report or certificate is to be included in the technical documentation where available.

IEC 60601-1-2

IEC 60601-1-2 is identified as the applicable collateral standard concerning electromagnetic compatibility of medical electrical equipment.

The standard is relevant to the electromagnetic compatibility of the **BN-100 LUX** and to the assessment of electromagnetic emissions and immunity applicable to the device.

The corresponding EMC test report is to be included in the technical documentation where available.

IEC 60601-1-8

IEC 60601-1-8 is referenced in the available manufacturer documentation in relation to alarm systems for medical electrical equipment.

This standard is relevant to the audible and visual alarm functions implemented in the **BN-100 LUX**, including safety-related alarm conditions identified in the manufacturer documentation.

ISO 14971

ISO 14971 is the reference standard for the application of risk management to medical devices.

The manufacturer's risk management documentation identifies hazards, hazardous situations, risks and corresponding risk-control measures associated with the device.

The available Risk Management Report is therefore considered as part of the risk-management documentation supporting the safety assessment of the device.

Relationship with the Device

The standards identified above are related to different aspects of the **BN-100 LUX**, including:

- basic electrical safety and essential performance;
- electromagnetic compatibility;
- alarm systems;
- risk management.

The application of these standards shall be understood in conjunction with the applicable manufacturer documentation, test reports and certificates included in the technical documentation.

Documentation Status

The available documentation identifies the standards relevant to the device, but the presence of a reference to a standard does not, by itself, constitute a complete conformity assessment.

Where a standard requires objective evidence of compliance, the applicable test report, certificate or other supporting document should be included in the technical documentation.

No test result or conformity statement is introduced in this section unless supported by the corresponding manufacturer documentation.

Documentary Basis

This section is based primarily on the following manufacturer documentation:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.2.5 Stability

General

For the **BN-100 LUX** Infant Radiant Warmer, stability is considered in relation to the ability of the device and its components to maintain their specified characteristics during storage, transportation and the intended service period.

As the **BN-100 LUX** is a medical electrical device and not a pharmaceutical product, stability is addressed through the manufacturer's documented storage conditions, service-life information, environmental limitations and component lifetime information.

Storage Conditions

The manufacturer documentation specifies the following storage conditions for the device:

- Storage temperature: −40–+55 °C;
- Relative humidity: not more than 95%;
- Atmospheric pressure: 500–1060 hPa.

These conditions define the environmental limits specified by the manufacturer for storage of the device when it is not in operation.

Operating Conditions

The manufacturer documentation specifies the following environmental conditions for operation:

- Ambient temperature: 18–30 °C;
- Relative humidity: 30–75%;
- Atmospheric pressure: 700–1060 hPa;
- Air velocity: less than 0.3 m/s.

These operating conditions are relevant to maintaining the documented performance and safety characteristics of the device during use.

Service Life

The manufacturer documentation specifies a theoretical service life of **8 years** for the device.

This service-life value represents the manufacturer's documented theoretical lifetime of the equipment under the specified conditions of use.

The actual service life may depend on the operating conditions, maintenance, replacement of components and continued compliance with the applicable safety and performance requirements.

Heater Tube Lifetime

The manufacturer documentation specifies a theoretical operating lifetime of approximately **25,000 hours** for the radiant heater tube.

The heater tube is a critical functional component of the radiant heating system and its documented lifetime is therefore relevant to the expected service period of the device.

Replacement of components, where applicable, shall be performed in accordance with the manufacturer's maintenance and service documentation.

Durability Considerations

The stability and durability of the **BN-100 LUX** are related to the continued integrity and performance of its electrical, mechanical, thermal and temperature-monitoring components.

The manufacturer documentation identifies critical components and safety functions that contribute to the safe operation of the finished device.

No additional durability period, accelerated ageing result or stability study is introduced in this section unless supported by the available manufacturer documentation.

Transportation and Storage

The documented storage environmental limits provide the available manufacturer basis for assessing the environmental conditions applicable when the device is not in operation.

The available documentation does not provide a separate accelerated ageing study or transportation stability study for the finished device.

Accordingly, no additional transportation or ageing performance claim is introduced in this section.

Assessment of Stability

Based on the available manufacturer documentation, the stability-related characteristics of the **BN-100 LUX** are documented primarily through:

- specified storage conditions;
- specified operating conditions;
- theoretical service life of the device;
- theoretical lifetime of the radiant heater tube;
- maintenance and replacement of applicable components.

These documented characteristics provide the available basis for addressing the stability and durability of the finished medical device within the technical documentation.

Documentation Limitation

The available manufacturer documentation does not provide a separate pharmaceutical-style stability study for the **BN-100 LUX**.

No such study is considered necessary as a pharmaceutical stability study because the product is a medical electrical device.

Any additional evidence concerning ageing, long-term durability, transportation validation or component replacement intervals should be supported by the corresponding manufacturer documentation where applicable.

Documentary Basis

This section is based primarily on the following manufacturer documentation:

- BN-100 / BN-100 LUX User Manual;
- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0.

3.3 Risk Analysis Report

3.3.1 Risk Analysis

General

Risk analysis for the **BN-100 LUX** Infant Radiant Warmer is addressed through the manufacturer's risk management documentation.

The available Risk Management Report identifies hazards and hazardous situations associated with the design, manufacture and use of the device and identifies corresponding risk-control measures.

The risk analysis is considered in relation to the intended use of the device, its operating principles, its electrical and thermal functions, its temperature monitoring system and its mechanical structure.

Risk Management Basis

The available manufacturer risk management documentation is identified as:

- **Risk Management Report (BN-100/BN-100A)**;
- Document No.: **YF-7.3-24-02 (w)**;
- Version: **A/0**.

The document identifies risks associated with the medical device and provides documented risk-control measures.

The reference to BN-100/BN-100A corresponds to the title of the available manufacturer risk management document. It is used here as documentary evidence and does not change the identification of the device subject to this dossier, which is the **BN-100 LUX**.

Identification of Hazards

The manufacturer risk management documentation considers hazards associated with several functional and component groups of the device.

These include, in particular:

- electrical hazards;

- thermal hazards;
- temperature-control hazards;
- temperature-sensor failures;
- alarm-system failures;
- mechanical hazards;
- hazards associated with patient-contact components;
- hazards associated with component or system failures;
- hazards associated with manufacturing and assembly processes.

The identified hazards are considered in relation to the intended use and reasonably foreseeable conditions of use of the device.

Electrical Hazards

The risk management documentation identifies electrical components and circuits relevant to the safety of the device.

These include components such as:

- power supply cord;
- input socket and fuse;
- relay;
- rocker switch;
- switching power supply;
- filter;
- internal electrical wiring;
- printed circuit board;
- auxiliary electrical components.

Potential risks associated with electrical component failure or abnormal electrical conditions are addressed through the manufacturer's documented risk-control measures.

Thermal Hazards

The principal function of the **BN-100 LUX** is to provide controlled radiant heat to newborn and infant patients.

Accordingly, excessive temperature, insufficient temperature control and failure of the heating system are relevant hazards considered in the risk management documentation.

The manufacturer identifies temperature monitoring, temperature control, over-temperature protection and alarm functions as safety-related measures.

Temperature Monitoring Hazards

The device uses a skin temperature sensor to monitor the patient's temperature.

The risk management documentation identifies failures affecting the skin temperature sensor, including:

- sensor short circuit;
- sensor open circuit;
- abnormal temperature measurement.

Corresponding alarm and monitoring functions are provided as documented risk-control measures.

Alarm-Related Hazards

The manufacturer identifies alarm functions associated with safety-related conditions of the device.

These include:

- over-temperature alarm;
- temperature deviation alarm;
- skin temperature sensor short-circuit alarm;
- skin temperature sensor open-circuit alarm;
- power failure alarm;
- manual-mode alarm.

The alarm functions are intended to provide an indication of specified abnormal or potentially hazardous conditions.

Mechanical Hazards

The risk analysis also considers mechanical characteristics of the device, including the patient support structure and associated components.

The manufacturer specifies mechanical limits for components such as the mattress support and tray.

These limits are relevant to the prevention of mechanical overload and associated hazards during intended operation.

Patient-Contact Hazards

The manufacturer identifies components that may come into direct contact with the patient, including:

- cotton baby mattress;
- stainless-steel skin temperature sensor probe.

These components are considered within the manufacturer's risk management documentation with regard to patient contact and associated biological considerations.

Manufacturing-Related Risks

The available risk management documentation also considers risks associated with manufacturing and assembly processes.

Critical manufacturing and assembly steps are considered in relation to the potential effect of process failures on the safety and performance of the finished device.

The corresponding controls are addressed within the manufacturer's documented production and risk-management framework.

Risk Analysis Approach

The risk analysis considers the relationship between:

- intended use;
- device functions;
- identified hazards;
- reasonably foreseeable hazardous situations;
- possible causes of hazardous situations;
- potential harm;
- implemented risk-control measures.

The available manufacturer Risk Management Report constitutes the primary documentary basis for this analysis.

Documentation Limitation

The present section summarizes the risk analysis documented by the manufacturer.

No new hazard, probability value, severity value or residual-risk classification is introduced independently of the available manufacturer risk management documentation.

Where detailed risk-analysis records are required, the original Risk Management Report and its associated risk-analysis tables should be provided as supporting technical documentation.

Documentary Basis

This section is based primarily on:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.3.2 Risk Evaluation

General

The risk evaluation of the **BN-100 LUX** Infant Radiant Warmer is addressed through the manufacturer's risk management documentation.

The evaluation considers the risks associated with the identified hazards and hazardous situations and their relationship to the intended use and safety-related functions of the device.

The available Risk Management Report provides the documentary basis for the risk evaluation.

Risk Evaluation Basis

The manufacturer's risk management documentation is identified as:

- **Risk Management Report (BN-100/BN-100A)**;
- Document No.: **YF-7.3-24-02 (w)**;
- Version: **A/0**.

The document title is retained as provided in the manufacturer documentation. The device covered by the present dossier remains the **BN-100 LUX**.

Evaluation of Identified Risks

The identified risks are evaluated according to their potential effect on the safety and performance of the device.

The risk evaluation takes into consideration the nature of the hazard, the reasonably foreseeable conditions that may lead to a hazardous situation and the potential harm to the patient, operator or other persons.

The manufacturer's risk management process provides the basis for determining whether the identified risks require risk-control measures.

Electrical Risk Evaluation

Electrical hazards are evaluated in relation to the electrical architecture and components of the device.

Relevant components include the power supply, input socket, fuse, relay, switching power supply, filter, internal wiring and printed circuit board.

The evaluation considers the possible consequences of electrical component failure or abnormal electrical conditions and their potential impact on patient and operator safety.

Thermal Risk Evaluation

Thermal risks are particularly relevant to the **BN-100 LUX** because the principal function of the device is to provide controlled radiant heat.

The evaluation therefore considers conditions such as:

- excessive heating;
- insufficient or uncontrolled heating;

- temperature-control failure;
- temperature-monitoring failure;
- deviation from the specified temperature condition.

These risks are considered together with the documented temperature-control and alarm functions of the device.

Temperature Sensor Risk Evaluation

The skin temperature sensor is a safety-related component of the temperature-monitoring system.

The manufacturer documentation identifies sensor-related failure conditions, including short circuit and open circuit.

These conditions are evaluated in relation to their potential effect on temperature monitoring and patient safety.

The corresponding alarm functions provide a documented means of detecting specified sensor failures.

Alarm System Risk Evaluation

The manufacturer identifies several alarm conditions associated with potentially hazardous operating conditions.

These include over-temperature, temperature deviation, sensor failure, power failure and manual-mode operation.

The evaluation of these conditions is related to the ability of the alarm system to provide an indication of the corresponding abnormal condition.

Mechanical Risk Evaluation

Mechanical risks are evaluated in relation to the structure and intended use of the patient support system and associated components.

The manufacturer's documented load limits for the mattress support and tray provide defined mechanical operating limits.

The crib inclination and adjustment mechanism are also considered in relation to the intended positioning function of the patient support.

Patient-Contact Risk Evaluation

The risk evaluation includes consideration of components that are in direct contact with the patient.

The manufacturer identifies the cotton baby mattress and stainless-steel skin temperature sensor probe as patient-contact components.

The corresponding biological considerations are addressed within the manufacturer's risk management documentation.

Manufacturing Risk Evaluation

The manufacturer's risk management documentation considers risks that may arise from manufacturing and assembly processes.

Such risks are evaluated according to their potential effect on the safety and performance of the finished device.

The corresponding process controls and manufacturing controls form part of the overall risk-control framework.

Risk Acceptability

The acceptability of identified risks is determined within the manufacturer's documented risk management process.

This section does not establish independent numerical acceptance criteria or introduce a new risk matrix.

Where detailed severity, probability, risk levels or acceptance decisions are required, the original manufacturer's Risk Management Report shall be used as the controlling documentary reference.

Overall Evaluation Approach

The risk evaluation considers the relationship between the identified hazards, hazardous situations, potential harms and the corresponding risk-control measures.

The evaluation is performed in the context of the intended use of the **BN-100 LUX** and its documented safety-related functions.

The available manufacturer documentation therefore provides the primary basis for the evaluation of the identified risks.

Documentation Limitation

The present section summarizes the risk evaluation documented by the manufacturer.

No independent severity, probability or risk-ranking values are introduced where they are not explicitly available in the supporting manufacturer documentation.

The original risk-analysis and risk-evaluation records should be retained as part of the technical documentation supporting this section.

Documentary Basis

This section is based primarily on:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.3.3 Risk Control

General

Risk control for the **BN-100 LUX** Infant Radiant Warmer is addressed through the risk-control measures documented by the manufacturer.

The identified control measures are related to the design, protective functions, monitoring systems, alarms, components and manufacturing processes of the device.

The available Risk Management Report constitutes the primary documentary basis for the risk-control measures presented in this section.

Risk Control Principles

The manufacturer implements risk-control measures through the design and operation of the device.

The documented controls include, where applicable:

- inherently safe design characteristics;
- protective measures;
- monitoring functions;
- alarm functions;
- electrical protection;
- thermal protection;
- temperature monitoring;
- mechanical limitations;
- manufacturing and process controls.

The risk-control measures are considered in relation to the identified hazards and hazardous situations.

Electrical Risk Controls

Electrical safety is addressed through the use of documented electrical components and protective elements.

The manufacturer identifies safety-related components including:

- dual fuse input socket;
- power supply cord;
- fuse;
- relay;
- rocker switch;
- switching power supply;
- filter;
- internal electrical wiring;
- printed circuit board.

These components contribute to the control of electrical hazards associated with the operation of the device.

Thermal Risk Controls

Thermal safety is addressed through temperature control, temperature monitoring and over-temperature protection.

The manufacturer documentation specifies:

- controlled temperature range of 32–38 °C;
- over-temperature alarm at the documented thresholds;
- heating cut-off under specified over-temperature conditions;
- temperature deviation alarm;
- skin temperature monitoring.

These functions provide documented controls against hazardous temperature conditions.

Temperature Sensor Controls

The skin temperature sensor is part of the temperature monitoring and control system.

The manufacturer documents alarm functions for sensor failure conditions, including:

- skin temperature sensor short circuit;
- skin temperature sensor open circuit.

These alarm functions provide a means of detecting specified sensor faults and contribute to the control of risks associated with incorrect or unavailable temperature measurements.

Alarm Controls

The device incorporates audible and visual alarm functions associated with specified abnormal conditions.

The documented alarm functions include:

- over-temperature alarm;
- temperature deviation alarm;
- skin temperature sensor short-circuit alarm;
- skin temperature sensor open-circuit alarm;
- power failure alarm;
- manual-mode alarm.

The alarms provide information to the operator concerning specified abnormal operating conditions.

Mechanical Risk Controls

Mechanical risks are controlled through defined mechanical characteristics and operating limits.

The manufacturer specifies limits for components including:

- mattress support;
- tray;
- crib inclination and adjustment.

The documented load limits provide defined conditions for the intended use of the corresponding components.

Patient-Contact Risk Controls

The manufacturer identifies patient-contact components within the risk management documentation.

These include:

- cotton baby mattress;
- stainless-steel skin temperature sensor probe.

The corresponding biological considerations are addressed through the manufacturer's risk management process.

No additional biological safety claim is introduced beyond the available manufacturer documentation.

Manufacturing and Process Controls

The manufacturer's risk management documentation considers manufacturing and assembly processes as part of the overall risk-control framework.

Controls are applied to relevant manufacturing steps and critical components to reduce the possibility that manufacturing defects could adversely affect the safety or performance of the finished device.

The available documentation does not provide a complete independent validation package for every manufacturing process.

Accordingly, no additional process-validation claim is introduced in this section.

Information for Safety

The user manual provides information necessary for the appropriate operation of the device, including operating conditions, temperature-control modes, alarm conditions and applicable limitations.

The provision of operating instructions and warnings forms part of the documented risk-control framework.

The operator is expected to use the device in accordance with the manufacturer's instructions.

Risk-Control Verification

The effectiveness of risk-control measures is addressed through the manufacturer's technical documentation and risk management process.

Where objective verification requires a test report, inspection record or other evidence, the corresponding manufacturer documentation should be included in the technical file.

No independent test result is introduced in this section where such evidence is not available in the submitted documentation.

Residual Risk

Following implementation of the documented risk-control measures, the remaining risks are considered within the manufacturer's overall risk management process.

The assessment of the remaining risks is addressed in Section 3.4.4, "Overall Residual Risk Evaluation".

Documentation Limitation

This section summarizes the risk-control measures documented by the manufacturer.

No new risk-control measure, acceptance criterion or performance claim is introduced independently of the available manufacturer documentation.

Documentary Basis

This section is based primarily on:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.3.4 Overall Residual Risk Evaluation

General

The overall residual risk evaluation for the **BN-100 LUX** Infant Radiant Warmer is addressed through the manufacturer's risk management documentation.

Residual risk refers to the risk remaining after the documented risk-control measures have been implemented.

The overall assessment considers the relationship between the identified hazards, the implemented risk-control measures and the risks remaining after application of those measures.

Risk Management Basis

The available manufacturer risk management document is identified as:

- **Risk Management Report (BN-100/BN-100A)**;
- Document No.: **YF-7.3-24-02 (w)**;
- Version: **A/0**.

The title and model designation of the source document are retained as provided by the manufacturer. The medical device covered by the present dossier remains the **BN-100 LUX**.

Residual Risks Following Risk Control

The manufacturer has implemented documented risk-control measures addressing the identified electrical, thermal, mechanical, monitoring and alarm-related hazards.

The principal documented controls include:

- electrical protective components;
- temperature monitoring;
- temperature control;
- over-temperature protection;
- heating cut-off;
- audible and visual alarms;
- sensor fault detection;
- power failure indication;
- mechanical operating limits;
- manufacturing and assembly controls.

After implementation of these measures, residual risks may remain as an inherent consequence of the intended use and operating principle of the device.

Thermal Residual Risk

The **BN-100 LUX** provides radiant heat directly to the newborn or infant.

Consequently, thermal risk remains an inherent consideration even when temperature control, monitoring and over-temperature protection are provided.

The documented temperature-control and alarm functions constitute the manufacturer's measures for controlling this risk.

The residual thermal risk is therefore considered within the manufacturer's overall risk management process.

Electrical Residual Risk

Electrical hazards may remain as residual risks associated with the use of medical electrical equipment.

The manufacturer addresses these risks through the documented electrical design, protective components and safety-related functions.

The remaining electrical risk is considered within the manufacturer's overall risk management framework.

Mechanical Residual Risk

Mechanical risks may remain during operation, positioning, adjustment or loading of the patient support structure and associated components.

The manufacturer specifies mechanical limits and operating conditions for these components.

Compliance with the specified operating limits and the manufacturer's instructions forms part of the control of the associated residual risks.

Temperature Monitoring Residual Risk

Failure or incorrect operation of the skin temperature monitoring system may result in residual risk.

The manufacturer addresses documented sensor failure conditions through specific alarm functions, including short-circuit and open-circuit detection.

The remaining risk associated with temperature monitoring is considered within the overall manufacturer's risk management process.

Alarm-Related Residual Risk

Alarm functions are intended to alert the operator to specified abnormal conditions.

A residual risk remains if an abnormal condition is not detected, if the alarm is not appropriately perceived by the operator, or if the device is not operated in accordance with the manufacturer's instructions.

The manufacturer documentation addresses alarm-related risks through the documented alarm functions and operating instructions.

Information for Safety

Information provided in the manufacturer's user documentation forms part of the overall risk-control strategy.

The operating instructions provide information concerning:

- intended operation;
- temperature-control modes;
- alarm conditions;
- environmental operating conditions;
- storage conditions;
- applicable operating limitations.

Correct use of the device in accordance with the manufacturer's instructions is therefore part of the management of residual risk.

Overall Residual Risk

The overall residual risk is evaluated by considering the risks remaining after implementation of the documented risk-control measures.

The available manufacturer documentation provides the basis for this evaluation through the identified hazards, risk-control measures and corresponding safety functions.

No independent numerical residual-risk value or new risk-ranking result is introduced in this section where such information is not explicitly available in the manufacturer's documentation.

Benefit–Risk Consideration

The **BN-100 LUX** is intended to provide controlled radiant thermal support to newborn and infant patients requiring thermal care.

The intended clinical function of the device is considered together with the risks identified and controlled through the manufacturer's risk management process.

The overall benefit–risk consideration shall be understood in conjunction with the manufacturer's intended-use information, risk management documentation and applicable technical evidence.

Documentation Limitation

The present section summarizes the overall residual-risk considerations supported by the available manufacturer documentation.

It does not replace the original manufacturer's risk management file and does not introduce independent residual-risk scores, acceptance criteria or benefit–risk conclusions that are not documented by the manufacturer.

Where the competent authority requires detailed residual-risk records, the original Risk Management Report and its associated risk-analysis tables should be included as supporting documentation.

Documentary Basis

This section is based primarily on:

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BN-100 / BN-100 LUX User Manual.

3.4 Annexes

3.4.1 General

The annexes constitute the supporting documentary material associated with the technical information presented in Part III of the technical dossier for the **BN-100 LUX** Infant Radiant Warmer.

The annexes provide documentary evidence supporting the device description, manufacturing information, specifications, safety information, risk management and other technical characteristics presented in the dossier.

Each supporting document shall retain its original title, document number, version and identification information as provided by the manufacturer or the relevant issuing organization.

3.4.2 Manufacturer Documentation

The manufacturer documentation included in the technical dossier may comprise the following documents, where available and applicable:

- BN-100 User Manual and applicable technical documentation;

- Risk Management Report (BN-100/BN-100A), Document No. YF-7.3-24-02 (w), Version A/0;
- BSI EU Quality Management System Certificate, Document No. MDR 796664 R000;
- ISO 13485 Quality Management System Certificate, Document No. MD 796665;
- Manufacturing process documentation;
- Raw material and component documentation;
- Finished product specifications;
- Applicable inspection and quality-control documentation;
- Maintenance and servicing documentation.

The original identification of each document shall be maintained in the submitted technical file.

3.4.3 Technical and Safety Documentation

The technical and safety annexes may include the following documents, where available and applicable:

- Electrical safety test reports;
- Electromagnetic compatibility test reports;
- Applicable alarm-system test documentation;
- Risk management documentation;
- Applicable certificates and declarations of conformity, including the manufacturer's EC Declaration of Conformity;
- Technical drawings and diagrams;
- Relevant component specifications.

Only documents available in the submission package and applicable to the **BN-100 LUX** shall be presented as supporting evidence.

3.4.4 Device Identification Documentation

Documents supporting device identification may include, where available and applicable:

- Device photographs;
- Identification labels;
- Model and serial-number information;
- Nameplate documentation;
- Packaging and identification information.

The identification information provided in the annexes shall be consistent with the device covered by the present dossier.

3.4.5 Risk Management Annexes

The manufacturer's detailed risk management records may be provided as supporting annexes to this section.

These records may include, where available:

- Hazard identification tables;
- Risk evaluation tables;
- Risk-control measures;
- Residual-risk evaluation;
- Risk-control verification records;
- Overall residual-risk assessment.

The original manufacturer risk-management records shall be retained without altering their original document identification.

3.4.6 Supporting Evidence

The annexes may contain additional evidence supporting specific statements made elsewhere in the technical dossier.

Such evidence may include manufacturer-issued technical documents, certificates, declarations, reports, drawings, photographs or other controlled records.

The applicability of each document to the device covered by this dossier shall be assessed before the document is relied upon as supporting evidence.

3.4.7 Document Control

Each annex should be identified, where applicable, by:

- Annex number;
- Document title;
- Document number;
- Version or revision;
- Date;
- Manufacturer or issuing organization.

Where a supporting document has no document number or version, its original identification shall be retained rather than creating an identification number that does not originate from the issuing organization.

3.4.8 Documentary Basis

The annexes are based on the technical and documentary evidence available for the **BN-100 LUX** Infant Radiant Warmer.

The final annex list shall correspond to the documents actually included in the submission package. Each listed document shall retain its original identification and shall be included only where available and applicable.

3.5 Bibliographical References

3.5.1 General

The bibliographical and documentary references listed in this section identify the principal technical and regulatory sources used to prepare Part III of the technical dossier for the **BN-100 LUX** Infant Radiant Warmer.

The references are limited to documents and standards relevant to the technical information presented in the dossier.

3.5.2 Manufacturer Documentation

1. **BN-100 User Manual**, manufacturer documentation, Version 2.2, as available in the technical documentation.
2. **Risk Management Report (BN-100/BN-100A)**, Document No. YF-7.3-24-02 (w), Version A/0.
3. **BSI EU Quality Management System Certificate**, Document No. MDR 796664 R000.
4. **ISO 13485 Quality Management System Certificate**, Document No. MD 796665.

The original title and identification of each manufacturer document or certificate shall be retained in the submitted technical file.

3.5.3 Applicable Standards and Reference Documents

The following standards are referenced in the available manufacturer documentation and are relevant to the technical and safety assessment of the device:

1. **IEC 60601-1** — Medical electrical equipment — Part 1: General requirements for basic safety and essential performance.
2. **IEC 60601-1-2** — Medical electrical equipment — Part 1-2: General requirements for basic safety and essential performance — Collateral Standard: Electromagnetic disturbances — Requirements and tests.
3. **IEC 60601-1-8** — Medical electrical equipment — Part 1-8: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems.
4. **ISO 14971** — Medical devices — Application of risk management to medical devices.

These references identify relevant technical standards but do not, by themselves, constitute evidence of conformity.

Where conformity to a particular standard is claimed, the corresponding manufacturer documentation, test report, certificate or other objective evidence should be provided where applicable.

3.5.4 Use of References

The references listed above support the technical description, safety assessment, risk management and specifications presented in Part III.

The manufacturer's documentation remains the primary source for device-specific characteristics and specifications.

Standards are used as technical reference documents and shall not be interpreted as evidence of compliance unless supported by the corresponding objective evidence.

3.5.5 Document Identification

For controlled manufacturer documents and certificates, the following information should be retained whenever available:

- Document title;
- Document number;
- Version or revision;
- Date;
- Manufacturer or issuing organization.

No document identification information shall be created where it is not provided by the original issuing organization.

3.5.6 Documentary Basis

This section is based on the manufacturer documentation, certificates and technical standards identified in the technical dossier for the **BN-100 LUX** Infant Radiant Warmer.

bsi.

By Royal Charter

EU Quality Management System Certificate

Regulation (EU) 2017/745, Annex IX Chapter I and III

MDR 796664 R000

Manufacturer: Zhengzhou Dison Instrument And Meter Co., Ltd.

Address:
No. 67, Jinsuo Road
High-tech Development Zone
Zhengzhou
Henan
450001
China

Single Registration Number: CN-MF-000021558

EU Authorised Representative: Share Info GmbH

Address:
Am Schulzentrum 12
41564 Kaarst
Germany

Scope: See attached **Device Schedule**

On the basis of our examination of the quality system in accordance with Regulation (EU) 2017/745, Annex IX Chapter I and III, the quality system meets the requirements of the Regulation. For the placing on the market of Class III devices, and Class IIb implantable devices that are not considered well-established technologies as specified in Article 52(4) an additional Annex IX Chapter II certificate is required.

For and on behalf of BSI, a Notified Body for the above Regulation (Notified Body Number 2797):

Graeme Tunbridge

Graeme Tunbridge, Senior Vice President Global Regulatory & Quality

First Issue Date: **2025-08-22**

Current Issue Date: **2025-08-22**

Starting Validity Date: **2025-08-22**

Expiry Date: **2030-08-21**

...making excellence a habit.™

Validity of this certificate is conditional on the Manufacturer's quality system being maintained to the requirements of the Regulation as demonstrated through the required surveillance activities of the Notified Body.
This certificate was issued electronically and is bound by the conditions of the contract.

NB Contact: BSI Group The Netherlands B.V., Say Building, John M. Keynesplein 9, 1066 EP, Amsterdam, Netherlands. Tel: + 31 (0) 20 346 07 80
Corporate Contact: BSI Group Assurance Limited, registered in England under number 05435540 at 389 Chiswick High Road, London, W4 4AL, UK.
A Member of the BSI Group of Companies.

A Member of the BSI Group of Companies.

bsi.

By Royal Charter

Certificate of Registration

QUALITY MANAGEMENT SYSTEM - ISO 13485:2016

This is to certify that: Zhengzhou Dison
Instrument and Meter Co., Ltd.
No. 67, Jinsuo Road
High-tech Development Zone
Zhengzhou
Henan
450001
China

郑州迪生仪器仪表有限公司
91410100614714585H
中国
河南省
郑州
高新技术产业开发区金梭路67号
邮编：450001

Holds Certificate No: **MD 796665**

and operates a Quality Management System which complies with the requirements of ISO 13485:2016 for the following scope:

The Design and Development, Manufacture and Distribution of Baby Incubators, Infant Radiant Warmers, Infant Phototherapy Units, Transport Infant Incubators and Reusable Pneumatic Pulmonary Resuscitators
婴儿培养箱、婴儿辐射保暖台、婴儿蓝光治疗仪、婴儿转运培养箱以及婴儿复苏器的设计、开发、制造和销售。

For and on behalf of BSI:

Graeme Tunbridge

Graeme Tunbridge, Senior Vice President Medical Devices

Original Registration Date: 2024-03-05
Latest Revision Date: 2024-03-05

Effective Date: 2024-03-05
Expiry Date: 2027-03-04

Page: 1 of 1

...making excellence a habit.

This certificate was issued electronically and remains the property of BSI and is bound by the conditions of contract.
An electronic certificate can be authenticated online.
Printed copies can be validated at www.bsi-global.com/ClientDirectory or telephone +86 10 8507 3000.

Information and Contact: BSI, John M. Keynesplein 9, 1066 EP Amsterdam The Netherlands. Tel: +31 (0) 20 3460 780
BSI Group The Netherlands B.V., registered in the Netherlands under number 33264284, at John M. Keynesplein 9, 1066 EP Amsterdam, The Netherlands
A Member of the BSI Group of Companies.

bsi.

By Royal Charter

EU Quality Management System Certificate

Regulation (EU) 2017/745, Annex IX Chapter I and III

MDR 796664 R000

Device Schedule: Class III and Class IIb devices

Class IIb	Intended Purpose
Baby Incubators	The baby incubators are used to provide a suitable environment for premature infants, sick infants and the normal infants leaving the mother's womb.
Transport Infant Incubators	The transport infant incubators provide a constant temperature environment for premature or weak neonate. The device is used for hospital transfer of an infant who needs intensive care.
Infant Radiant Warmers	The infant radiant warmers are mainly used for rescue, nursing and pediatric surgical treatment of newborns.

Device Schedule: Class IIa, Custom-made and other devices

Device(s)	Risk Classification
Infant Phototherapy Units	Class IIa

First Issue Date: **2025-08-22**

Current Issue Date: **2025-08-22**

Starting Validity Date: **2025-08-22**

Expiry Date: **2030-08-21**

...making excellence a habit.

Validity of this certificate is conditional on the Manufacturer's quality system being maintained to the requirements of the Regulation as demonstrated through the required surveillance activities of the Notified Body.
This certificate was issued electronically and is bound by the conditions of the contract.

NB Contact: BSI Group The Netherlands B.V., Say Building, John M. Keynesplein 9, 1066 EP, Amsterdam, Netherlands. Tel: + 31 (0) 20 346 07 80
Corporate Contact: BSI Group Assurance Limited, registered in England under number 05435540 at 389 Chiswick High Road, London, W4 4AL, UK.
A Member of the BSI Group of Companies.

bsi.

By Royal Charter

EU Quality Management System Certificate

Regulation (EU) 2017/745, Annex IX Chapter I and III

MDR 796664 R000

Certificate History

(References to applicable Common Specifications, Harmonized Standards complied with, and the relevant test and audit reports that support any of the below certificate changes may be requested from Certificate.Verification@bsigroup.com)

Date	Reference Number	Action
Current	30007263	Issued

First Issue Date: **2025-08-22**

Current Issue Date: **2025-08-22**

Starting Validity Date: **2025-08-22**

Expiry Date: **2030-08-21**

...making excellence a habit.™

Validity of this certificate is conditional on the Manufacturer's quality system being maintained to the requirements of the Regulation as demonstrated through the required surveillance activities of the Notified Body.
This certificate was issued electronically and is bound by the conditions of the contract.

NB Contact: BSI Group The Netherlands B.V., Say Building, John M. Keynesplein 9, 1066 EP, Amsterdam, Netherlands. Tel: + 31 (0) 20 346 07 80
Corporate Contact: BSI Group Assurance Limited, registered in England under number 05435540 at 389 Chiswick High Road, London, W4 4AL, UK.
A Member of the BSI Group of Companies.

中华人民共和国
PEOPLE'S REPUBLIC OF CHINA
医疗器械产品出口销售证明
CERTIFICATE FOR EXPORTATION OF MEDICAL PRODUCTS

证书编号:豫郑药监械出20240060
Certificate NO.: 豫郑药监械出20240060

产品名称: 见附页
Product(s): see attachment

规格型号: 见附页
Model: see attachment

产品注册或备案凭证号: 见附页
Registration certificate(s): see attachment

生产企业:郑州迪生仪器仪表有限公司
Manufacturer: ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD

生产企业住所: 郑州高新技术产业开发区金梭路67号
Address of manufacturer: NO.67 JINSUO ROAD,HIGH-TECH DEVELOPMENT ZONE,ZHENGZHOU,HENAN,CHINA

生产许可或备案凭证号: 豫食药监械生产许20160103号
Manufacturing License(s): 豫食药监械生产许20160103号

兹证明上述产品已准许在中国生产和销售。
This is to certify that the above products have been registered to be manufactured and sold in China.

证明有效日期至: 2026年06月14日
This certification valid until: 06/14/2026(M/D/Y)

备注:
Remark:

二〇二四年八月三十日
Henan Medical Products Administration

豫郑药监械出20240060附页

序号	产品名称	英文名称	规格型号（中文）	规格型号（英文）	产品注册证号或备案凭证号	单位
1	婴儿培养箱	BABY INCUBATOR	BB-100，BB-200，BB-300	BB-100，BB-200，BB-300	国械注准20143082260	郑州迪生仪器仪表有限公司 ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD
2	婴儿辐射保暖台	INFANT RADIANT WARMER	BN-100，BN-100A，BN-200	BN-100，BN-100A，BN-200	国械注准20193081987	郑州迪生仪器仪表有限公司 ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD
3	婴儿蓝光治疗仪	INFANT PHOTOTHERAPY UNIT	BL-30D，BL-50D，BL-60D，BL-100D，BL-200D，BL-360D	BL-30D，BL-50D，BL-60D，BL-100D，BL-200D，BL-360D	豫械注准20222090074	郑州迪生仪器仪表有限公司 ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD
4	婴儿转运培养箱	TRANSPORT INFANT INCUBATOR	BT-100	BT-100	国械注准20243081440	郑州迪生仪器仪表有限公司 ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD

EC Declaration of Conformity

Manufacturer:

Zhengzhou Dison Instrument and Meter Co.,Ltd
No.67 Jinsuo Road, High-Tech Development Zone, Zhengzhou, 450001, Henan, China

whose single Authorized Representative:

Share Info GmbH
Am Schulzentrum 12, 41564 Kaarst, Germany

We, the manufacturer, herewith declare that the products

Infant Radiant Warmers

Model: BN-100;BN-100A

(including system components and accessories)

Product Model	UDI-DI Code	Basic UDI-DI Code	EMDN Code
BN-100	06974805202041	697480520BN100BU	Z12080407
BN-100A	06974805202058	697480520BN100A8Y	

meet the Regulation (EU) 2017/745 which apply to them.

The medical device has been assigned to class IIb according to Rule 9 in Annex VIII of the Regulation (EU) 2017/745.

It bears the mark CE 2797

The product concerned has been manufactured under a quality management system according to Annex IX excluding chapter II of Regulation (EU) 2017/745.

Compliance of the designated product with the Regulation (EU) 2017/745 has been assessed and certified by the Notified Body:

BSI Group The Netherlands B.V.

Certificate No.: MDR 796664

Issue date:2025-08-22

Expiry date: 2030-08-21

This declaration of conformity is valid in connection with the release document for the respective batch of produced devices.

The above mentioned declaration of conformity is exclusively under the responsibility of

Zhengzhou Dison Instrument and Meter Co.,Ltd

Zhengzhou 2025.8.22 Xm Xm Management Representative

Place , date *Legally binding signature, Function*

EC declaration of conformity

DS-TCF-6.16-02(w), A/2

EC Declaration of Conformity

Manufacturer

Zhengzhou Dison Instrument and Meter Co.,Ltd
No.67 Jinsuo Road, High-Tech Development Zone, Zhengzhou, 450001, Henan, China

whose single Authorized Representative:

Share Info GmbH
Am Schulzentrum 12, 41564 Kaarst, Germany

We, the manufacturer, herewith declare that the products

Baby Incubators

Model: BB-100;BB-200;BB-300

(including system components and accessories)

Product Model	UDI-DI Code	Basic UDI-DI Code	EMDN Code
BB-100	06974805202003	697480520BB10098	Z1208040301
BB-200	06974805202010	697480520BB2009D	
BB-300	06974805202027	697480520BB3009J	

meet the Regulation (EU) 2017/745 which apply to them.

The medical device has been assigned to class IIb according to Rule 9 in Annex VIII of the Regulation (EU) 2017/745.

It bears the mark CE 2797

The product concerned has been manufactured under a quality management system according to Annex IX excluding chapter II of Regulation (EU) 2017/745.

Compliance of the designated product with the Regulation (EU) 2017/745 has been assessed and certified by the Notified Body:

BSI Group The Netherlands B.V.

Certificate No.: MDR 796664

Issue date:2025-08-22

Expiry date: 2030-08-21

This declaration of conformity is valid in connection with the release document for the respective batch of produced devices.

The above mentioned declaration of conformity is exclusively under the responsibility of

Zhengzhou Dison Instrument and Meter Co.,Ltd

Zhengzhou 2025.8.29 — Management Representative

Place , date — *Legally binding signature, Function*

EC declaration of conformity

DS-TCF-6.16-01(w), A/2

disonmed ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD

SARL SKALBEL SANTE
179 RUE HAMID KABLADJ HAMMAMAT
ALGER ALGERIE

ORIGINAL CONFORMITY CERTIFICATE

Date: Mar. 6th, 2026

US: ZHENGZHOU DISON INSTRUMENT AND METER CO., LTD

Let us declare under our responsibility that the following parts:

NEONATOLOGIE EQUIPEMENT

delivered according to the Invoice No.DS-WX2406-002B on Mar 6th ,2026 are in conformity with our specifications techniques and with all the check operations and of test carried out following the plans and European standards into force.

disonmed ZHENGZHOU DISON INSTRUMENT AND METER CO.,LTD

FREE SALE CERTIFICATE IN THE CONTRY OF ORIGIN

The competent authority:
attests, to the request of: SARL SKALBEL SANTE
179 RUE HAMID KABLADJ HAMMAMAT
ALGER ALGERIE

that the product: NEONATOLOGIE EQUIPEMENT
failing under tariff positions: 9018909919
manufactured by: Zhengzhou Dison Instrument And Meter Co., Ltd
are in compliance with current regulations and following international standards for safety requirements and consumer protection.
Is (are)Marketed in(s): PEOPLE'S REPUBLIC OF CHINA(P.R.C)
Proforma invoice no: DS-WX2406-002B DATE: Mar 6th ,2026